Supplemental Table and Figure Captions–

Table S1: Bayesian model outputs for spatial and temporal variable effects on insect herbivory categories on early-season leaves (Model Set 2). For each herbivory category and spatial model combination, the table shows: slope coefficient estimates (posterior means), standard errors (SE), 95% credible intervals (95% CI), whether the 95% CI excludes zero (Excludes 0?), 80% credible intervals (80% CI), whether the 80% CI excludes zero (Excludes 0?), and adjusted R^2 values (R^2 (Adj)). Each herbivory category was modeled separately using either latitude or longitude as the spatial predictor, alongside year and day of year as temporal covariates. Herbivory metrics span multiple damage categories including frequencies, richness measures, and percent area damaged (Figure 1).

Table S2: Bayesian model outputs for spatial and temporal variable effects on insect herbivory categories on late-season leaves (Model Set 2). For each herbivory category and spatial model combination, the table shows: slope coefficient estimates (posterior means), standard errors (SE), 95% credible intervals (95% CI), whether the 95% CI excludes zero (Excludes 0?), 80% credible intervals (80% CI), whether the 80% CI excludes zero (Excludes 0?), and adjusted R^2 values (R^2 (Adj)). Each herbivory category was modeled separately using either latitude or longitude as the spatial predictor, alongside year and day of year as temporal covariates. Herbivory metrics span multiple damage categories including frequencies, richness measures, and percent area damaged (Figure 1).

Table S3: Bayesian model outputs for temporal variable effects on insect herbivory categories on early-season leaves (Model Set 1). For each herbivory category, the table shows: slope coefficient estimates (posterior means), standard errors (SE), 95% credible intervals (95% CI), whether the 95% CI excludes zero (Excludes 0?), 80% credible intervals (80% CI), whether the 80% CI excludes zero (Excludes 0?), and adjusted R^2 values (R^2 (Adj)). Temporal predictors include year, day of year, and their interaction (Year × Day of Year). Herbivory metrics span multiple damage categories including frequencies, richness measures, and percent area damaged (Figure 1).

Table S4: Bayesian model outputs for temporal variable effects on insect herbivory categories on late-season leaves (Model Set 1). For each herbivory category, the table shows: slope coefficient estimates (posterior means), standard errors (SE), 95% credible intervals (95% CI), whether the 95% CI excludes zero (Excludes 0?), 80% credible intervals (80% CI), whether the 80% CI excludes zero (Excludes 0?), and adjusted R^2 values (R^2 (Adj)). Temporal predictors include year, day of year, and their interaction (Year × Day of Year). Herbivory metrics span multiple damage categories including frequencies, richness measures, and percent area damaged (Figure 1).

Table S5: Pearson correlations between collection year and the annual climate variables included in Model Set 3 for late season and early-season leaves. Pearson r and associated p-values reflect the strength and significance of the linear relationship between year and each climate variable listed. Significance codes: $^{**}p < 0.01$, $^{*}p < 0.05$, ns = not significant.

Table S6: Bayesian model outputs for climate variable effects on insect herbivory categories on early-season leaves (Model Set 3). For each herbivory category and climate variable combination, the table shows: slope coefficient estimates (posterior means), standard errors (SE), 95% credible intervals (95% CI), whether the 95% CI excludes zero (Excludes 0?), 80% credible intervals (80% CI), whether the 80% CI excludes zero (Excludes 0?), and adjusted R^2 values (R^2 (Adj)). Climate variables include: annual heat moisture index (AHM), extreme maximum temperature (EXT), mean annual radiation (MAR), May–September precipitation (MSP), precipitation as snow (PAS), and summer heat moisture index (SHM). Herbivory metrics span multiple damage categories including frequencies, richness measures, and percent area damaged (Figure 1).

Table S7: Bayesian model outputs for climate variable effects on insect herbivory categories on late-season leaves (Model Set 3). For each herbivory category and climate variable combination, the table shows: slope coefficient estimates (posterior means), standard errors (SE), 95% credible intervals (95% CI), whether the 95% CI excludes zero (Excludes 0?), 80% credible intervals (80% CI), whether the 80% CI excludes zero (Excludes 0?), and adjusted R^2 values (R^2 (Adj)). Climate variables include: annual heat moisture index (AHM), extreme maximum temperature (EXT), mean annual precipitation (MAP), mean annual radiation (MAR), May–September precipitation (MSP), relative humidity (RH), summer heat moisture index (SHM), and temperature difference (TD). Herbivory metrics span multiple damage categories including frequencies, richness measures, and percent area damaged (Figure 1).

Figure S1: Proportion of insect herbivory across decades for early-season (top) and late-season leaves (bottom). Total (teal), chewing (orange), generalized (purple), specialized (pink), mine (green), and gall (yellow) damage are shown across decades. Missing bars indicate an absence of that functional feeding group from within the herbarium sheets and decade. The number of unique herbarium specimens per decade and relative leaf age are provided at the top of each decadal bin (n = X).

Figure S2: Histogram showing the distribution of early-season (A) and late-season (B) leaves through months. Leaves were classified based on morphology and the day of year (DOY) collected. Grey bars represent leaves that were thrown out of the dataset, while light green represents early-season leaves, and dark green represents late-season leaves. Because *Quercus lobata* is polycyclism, multiple flushes of leaves throughout a growing season, leaves that flushed later in the season (ex, a leaf from August that is morphologically young) are exposed to

a different herbivore community. Because of this, that leaf was reclassified as late-season (dark green; A). Dashed lines represent the intervals that herbivory on leaves was constrained to (A, early-season = April - July; B, late-season = August - December).

Figure S3: Correlation heat map for space and time predictor variables for early- (A) and late-season (B) leaves. The darker the colored numbers, the stronger the correlation with red representing positive correlations and blue representing negative correlations.

Table S1:

Time/Space Model outputs - Early-Season Leaves

Variable	Slope	SE	95% CI	Excludes 0?	80% CI	Excludes 0?	R^2
Chewing Frequency							
Latitude Model							
Year	0.0510	0.1066	[-0.1548, 0.2575]	No	[-0.0844, 0.1858]	No	0.029
Day of Year	-0.1019	0.1052	[-0.3090, 0.1042]	No	[-0.2355, 0.0327]	No	0.029
Latitude	0.0803	0.1080	[-0.1315, 0.2914]	No	[-0.0572, 0.2195]	No	0.029
Longitude Model							
Year	0.0486	0.1065	[-0.1602, 0.2590]	No	[-0.0882, 0.1861]	No	0.033
Day of Year	-0.1093	0.1026	[-0.3054, 0.0963]	No	[-0.2394, 0.0219]	No	0.033
Longitude	-0.1751	0.1039	[-0.3820, 0.0241]	No	[-0.3077, -0.0421]	Yes	0.033
Gall Frequency							
Latitude Model							
Year	0.0149	0.0819	[-0.1440, 0.1781]	No	[-0.0902, 0.1181]	No	0.015
Day of Year	0.0104	0.0832	[-0.1516, 0.1735]	No	[-0.0957, 0.1168]	No	0.015
Latitude	0.0113	0.0810	[-0.1487, 0.1686]	No	[-0.0915, 0.1154]	No	0.015
Longitude Model							
Year	0.0139	0.0821	[-0.1417, 0.1766]	No	[-0.0898, 0.1194]	No	0.015
Day of Year	0.0098	0.0808	[-0.1501, 0.1651]	No	[-0.0946, 0.1131]	No	0.015
Longitude	-0.0279	0.0801	[-0.1833, 0.1293]	No	[-0.1298, 0.0756]	No	0.015
Gall Richness							
Latitude Model							
Year	0.2918	0.7215	[-1.1962, 1.6946]	No	[-0.6018, 1.1858]	No	0.199
Day of Year	0.1159	0.6035	[-1.1206, 1.2954]	No	[-0.6461, 0.8672]	No	0.199
Latitude	0.2574	0.6083	[-0.9282, 1.4911]	No	[-0.5007, 1.0271]	No	0.199
Longitude Model							
Year	0.2705	0.7149	[-1.2294, 1.6797]	No	[-0.5959, 1.1453]	No	0.199
Day of Year	0.1535	0.5942	[-1.0430, 1.3412]	No	[-0.5916, 0.9099]	No	0.199
Longitude	-0.6212	0.6409	[-1.9029, 0.6663]	No	[-1.4133, 0.1572]	No	0.199
Generalized Frequency							
Latitude Model							
Year	0.0027	0.0853	[-0.1662, 0.1693]	No	[-0.1062, 0.1117]	No	0.015
Day of Year	-0.0122	0.0851	[-0.1813, 0.1521]	No	[-0.1217, 0.0965]	No	0.015
Latitude	0.0018	0.0866	[-0.1682, 0.1705]	No	[-0.1115, 0.1118]	No	0.015
Longitude Model							
Year	0.0044	0.0846	[-0.1602, 0.1713]	No	[-0.1044, 0.1118]	No	0.015
Day of Year	-0.0134	0.0860	[-0.1830, 0.1585]	No	[-0.1220, 0.0966]	No	0.015
Longitude	0.0034	0.0864	[-0.1673, 0.1741]	No	[-0.1072, 0.1136]	No	0.015
Generalized Richness							
Latitude Model							
Year	0.0445	0.2591	[-0.4583, 0.5650]	No	[-0.2824, 0.3745]	No	0.075
Day of Year	-0.0450	0.2652	[-0.5791, 0.4689]	No	[-0.3805, 0.2908]	No	0.075
Latitude	0.0445	0.2638	[-0.4758, 0.5711]	No	[-0.2860, 0.3770]	No	0.075
Longitude Model							
Year	0.0420	0.2600	[-0.4567, 0.5527]	No	[-0.2819, 0.3795]	No	0.078
Day of Year	-0.0507	0.2677	[-0.5941, 0.4663]	No	[-0.3963, 0.2801]	No	0.078
Longitude	-0.1911	0.2602	[-0.7015, 0.3221]	No	[-0.5261, 0.1398]	No	0.078
Mine Frequency							
Latitude Model							
Year	0.0124	0.0913	[-0.1671, 0.1902]	No	[-0.1040, 0.1293]	No	0.019
Day of Year	0.0406	0.0916	[-0.1421, 0.2192]	No	[-0.0771, 0.1581]	No	0.019
Latitude	0.0797	0.0933	[-0.1021, 0.2611]	No	[-0.0407, 0.1996]	No	0.019
Longitude Model							
Year	0.0167	0.0919	[-0.1614, 0.1985]	No	[-0.1002, 0.1350]	No	0.019
Day of Year	0.0378	0.0934	[-0.1465, 0.2247]	No	[-0.0811, 0.1581]	No	0.019
Longitude	-0.0809	0.0905	[-0.2586, 0.0945]	No	[-0.1973, 0.0350]	No	0.019
Mine Richness							
Latitude Model							
Year	0.0850	0.2837	[-0.4465, 0.6742]	No	[-0.2675, 0.4513]	No	0.094
Day of Year	0.1959	0.2715	[-0.3420, 0.7277]	No	[-0.1440, 0.5416]	No	0.094
Latitude	0.4151	0.2895	[-0.1282, 1.0113]	No	[0.0516, 0.7901]	Yes	0.094
Longitude Model							
Year	0.1648	0.2930	[-0.3795, 0.7756]	No	[-0.1976, 0.5389]	No	0.101
Day of Year	0.1977	0.2777	[-0.3443, 0.7456]	No	[-0.1588, 0.5498]	No	0.101
Longitude	-0.5059	0.3025	[-1.1430, 0.0562]	No	[-0.8959, -0.1314]	Yes	0.101
% Area Chewed							
Latitude Model							
Year	0.0051	0.0728	[-0.1385, 0.1485]	No	[-0.0889, 0.0979]	No	0.006
Day of Year	-0.0696	0.0755	[-0.2207, 0.0735]	No	[-0.1665, 0.0254]	No	0.006
Latitude	0.0623	0.0797	[-0.0924, 0.2221]	No	[-0.0400, 0.1641]	No	0.006
Longitude Model							
Year	0.0072	0.0724	[-0.1310, 0.1504]	No	[-0.0845, 0.1006]	No	0.007
Day of Year	-0.0769	0.0762	[-0.2284, 0.0713]	No	[-0.1739, 0.0210]	No	0.007
Longitude	-0.1029	0.0755	[-0.2512, 0.0424]	No	[-0.1995, -0.0063]	Yes	0.007
% Area Galled							
Latitude Model							
Year	0.0158	0.0607	[-0.1003, 0.1391]	No	[-0.0607, 0.0932]	No	0.006
Day of Year	0.0559	0.0608	[-0.0623, 0.1753]	No	[-0.0227, 0.1339]	No	0.006
Latitude	0.0243	0.0588	[-0.0904, 0.1393]	No	[-0.0508, 0.1005]	No	0.006
Longitude Model							
Year	0.0188	0.0604	[-0.0996, 0.1394]	No	[-0.0566, 0.0965]	No	0.006
Day of Year	0.0551	0.0600	[-0.0629, 0.1704]	No	[-0.0209, 0.1312]	No	0.006
Longitude	-0.0132	0.0585	[-0.1262, 0.1000]	No	[-0.0885, 0.0614]	No	0.006
% Area Mined							
Latitude Model							
Year	0.0292	0.0727	[-0.1111, 0.1706]	No	[-0.0634, 0.1242]	No	0.001
Day of Year	0.0185	0.0700	[-0.1196, 0.1546]	No	[-0.0715, 0.1090]	No	0.001
Latitude	0.0225	0.0692	[-0.1116, 0.1613]	No	[-0.0662, 0.1113]	No	0.001
Longitude Model							
Year	0.0309	0.0709	[-0.1071, 0.1706]	No	[-0.0596, 0.1222]	No	0.001
Day of Year	0.0168	0.0701	[-0.1255, 0.1518]	No	[-0.0740, 0.1063]	No	0.001
Longitude	-0.0307	0.0701	[-0.1700, 0.1068]	No	[-0.1199, 0.0588]	No	0.001
Specialized Frequency							
Latitude Model							
Year	0.0820	0.1051	[-0.1181, 0.2865]	No	[-0.0517, 0.2146]	No	0.032
Day of Year	-0.0150	0.1027	[-0.2176, 0.1845]	No	[-0.1464, 0.1184]	No	0.032
Latitude	0.1442	0.1086	[-0.0663, 0.3567]	No	[0.0040, 0.2847]	Yes	0.032
Longitude Model							
Year	0.0795	0.1038	[-0.1260, 0.2852]	No	[-0.0520, 0.2106]	No	0.036
Day of Year	-0.0268	0.1031	[-0.2331, 0.1745]	No	[-0.1564, 0.1044]	No	0.036
Longitude	-0.2263	0.1066	[-0.4353, -0.0166]	Yes	[-0.3620, -0.0876]	Yes	0.036
Specialized Richness							
Latitude Model							
Year	0.1626	0.2347	[-0.2997, 0.6334]	No	[-0.1299, 0.4615]	No	0.249
Day of Year	-0.0138	0.2219	[-0.4499, 0.4423]	No	[-0.2899, 0.2680]	No	0.249
Latitude	0.2175	0.2142	[-0.1993, 0.6452]	No	[-0.0513, 0.4883]	No	0.249
Longitude Model							
Year	0.1450	0.2291	[-0.3147, 0.5979]	No	[-0.1380, 0.4356]	No	0.251
Day of Year	-0.0318	0.2168	[-0.4505, 0.3979]	No	[-0.3034, 0.2494]	No	0.251
Longitude	-0.5424	0.2312	[-1.0409, -0.1170]	Yes	[-0.8379, -0.2602]	Yes	0.251
All Frequency							
Latitude Model							
Year	0.0825	0.1096	[-0.1304, 0.2966]	No	[-0.0573, 0.2235]	No	0.031
Day of Year	-0.0254	0.1056	[-0.2409, 0.1773]	No	[-0.1581, 0.1090]	No	0.031
Latitude	0.1458	0.1097	[-0.0679, 0.3600]	No	[0.0069, 0.2847]	Yes	0.031
Longitude Model							
Year	0.0839	0.1058	[-0.1260, 0.2945]	No	[-0.0514, 0.2171]	No	0.034
Day of Year	-0.0343	0.1051	[-0.2415, 0.1711]	No	[-0.1682, 0.1007]	No	0.034
Longitude	-0.2243	0.1049	[-0.4358, -0.0212]	Yes	[-0.3601, -0.0909]	Yes	0.034
All Richness							
Latitude Model							
Year	0.1131	0.2102	[-0.3006, 0.5259]	No	[-0.1531, 0.3777]	No	0.227
Day of Year	-0.0380	0.1995	[-0.4301, 0.3648]	No	[-0.2893, 0.2119]	No	0.227
Latitude	0.1920	0.2023	[-0.2018, 0.6095]	No	[-0.0568, 0.4429]	No	0.227
Longitude Model							
Year	0.1008	0.2041	[-0.3095, 0.4952]	No	[-0.1578, 0.3568]	No	0.229
Day of Year	-0.0550	0.1939	[-0.4412, 0.3314]	No	[-0.2942, 0.1922]	No	0.229
Longitude	-0.4785	0.2006	[-0.8900, -0.0969]	Yes	[-0.7318, -0.2316]	Yes	0.229

Table S2:

Time/Space Model outputs - Late-Season Leaves

Variable	Slope	SE	95% CI	Excludes 0?	80% CI	Excludes 0?	R^2
Chewing Frequency							
Latitude Model							
Year	0.0398	0.1158	[-0.1997, 0.2644]	No	[-0.1092, 0.1844]	No	0.030
Day of Year	0.0682	0.1024	[-0.1308, 0.2715]	No	[-0.0606, 0.1993]	No	0.030
Latitude	0.1055	0.1121	[-0.1166, 0.3252]	No	[-0.0365, 0.2477]	No	0.030
Longitude Model							
Year	0.0400	0.1253	[-0.2067, 0.2816]	No	[-0.1179, 0.2012]	No	0.030
Day of Year	0.0658	0.1030	[-0.1349, 0.2719]	No	[-0.0646, 0.1958]	No	0.030
Longitude	-0.0892	0.1221	[-0.3272, 0.1527]	No	[-0.2446, 0.0664]	No	0.030
Gall Frequency							
Latitude Model							
Year	0.1215	0.0956	[-0.0696, 0.3077]	No	[0.0005, 0.2441]	Yes	0.020
Day of Year	0.0499	0.0867	[-0.1202, 0.2180]	No	[-0.0602, 0.1612]	No	0.020
Latitude	0.0802	0.0958	[-0.1062, 0.2669]	No	[-0.0426, 0.2035]	No	0.020
Longitude Model							
Year	0.1048	0.1049	[-0.1051, 0.3092]	No	[-0.0267, 0.2388]	No	0.019
Day of Year	0.0455	0.0861	[-0.1214, 0.2154]	No	[-0.0639, 0.1557]	No	0.019
Longitude	-0.0359	0.1033	[-0.2377, 0.1671]	No	[-0.1667, 0.0947]	No	0.019
Gall Richness							
Latitude Model							
Year	0.2138	0.2397	[-0.2619, 0.6941]	No	[-0.0770, 0.5116]	No	0.105
Day of Year	0.3505	0.3082	[-0.2088, 0.9838]	No	[-0.0289, 0.7545]	No	0.105
Latitude	0.2104	0.2317	[-0.2533, 0.6834]	No	[-0.0751, 0.5011]	No	0.105
Longitude Model							
Year	0.1776	0.2662	[-0.3528, 0.7020]	No	[-0.1566, 0.5058]	No	0.105
Day of Year	0.3178	0.3057	[-0.2505, 0.9584]	No	[-0.0589, 0.7117]	No	0.105
Longitude	-0.0563	0.2548	[-0.5676, 0.4434]	No	[-0.3760, 0.2623]	No	0.105
Generalized Frequency							
Latitude Model							
Year	-0.0214	0.0880	[-0.1982, 0.1511]	No	[-0.1357, 0.0912]	No	0.014
Day of Year	0.0409	0.0798	[-0.1159, 0.2023]	No	[-0.0591, 0.1440]	No	0.014
Latitude	-0.0489	0.0880	[-0.2213, 0.1265]	No	[-0.1604, 0.0634]	No	0.014
Longitude Model							
Year	-0.0162	0.0962	[-0.2067, 0.1727]	No	[-0.1371, 0.1057]	No	0.013
Day of Year	0.0420	0.0795	[-0.1077, 0.2020]	No	[-0.0588, 0.1443]	No	0.013
Longitude	0.0316	0.0931	[-0.1530, 0.2115]	No	[-0.0900, 0.1491]	No	0.013
Generalized Richness							
Latitude Model							
Year	0.2785	0.2079	[-0.1146, 0.7076]	No	[0.0251, 0.5436]	Yes	0.141
Day of Year	0.5142	0.2919	[0.0080, 1.1547]	Yes	[0.1639, 0.8935]	Yes	0.141
Latitude	0.1372	0.2013	[-0.2677, 0.5377]	No	[-0.1175, 0.3903]	No	0.141
Longitude Model							
Year	0.2231	0.2284	[-0.2244, 0.6735]	No	[-0.0643, 0.5085]	No	0.147
Day of Year	0.4864	0.2932	[-0.0186, 1.1292]	No	[0.1306, 0.8753]	Yes	0.147
Longitude	0.0379	0.2179	[-0.3781, 0.4875]	No	[-0.2341, 0.3116]	No	0.147
Mine Frequency							
Latitude Model							
Year	0.0759	0.1045	[-0.1313, 0.2822]	No	[-0.0558, 0.2098]	No	0.019
Day of Year	0.0361	0.0899	[-0.1393, 0.2143]	No	[-0.0771, 0.1520]	No	0.019
Latitude	0.1157	0.1012	[-0.0820, 0.3145]	No	[-0.0151, 0.2434]	No	0.019
Longitude Model							
Year	0.0931	0.1107	[-0.1204, 0.3103]	No	[-0.0484, 0.2320]	No	0.020
Day of Year	0.0325	0.0904	[-0.1453, 0.2138]	No	[-0.0823, 0.1464]	No	0.020
Longitude	-0.1274	0.1067	[-0.3356, 0.0788]	No	[-0.2635, 0.0090]	No	0.020
Mine Richness							
Latitude Model							
Year	0.0134	0.1832	[-0.3677, 0.3620]	No	[-0.2205, 0.2399]	No	0.099
Day of Year	0.1237	0.1984	[-0.2391, 0.5328]	No	[-0.1172, 0.3796]	No	0.099
Latitude	0.2861	0.1923	[-0.0756, 0.6754]	No	[0.0414, 0.5328]	Yes	0.099
Longitude Model							
Year	0.0468	0.1957	[-0.3587, 0.4161]	No	[-0.2080, 0.2878]	No	0.096
Day of Year	0.1151	0.1950	[-0.2299, 0.5327]	No	[-0.1204, 0.3696]	No	0.096
Longitude	-0.2844	0.2081	[-0.7076, 0.1147]	No	[-0.5486, -0.0238]	Yes	0.096
% Area Chewed							
Latitude Model							
Year	0.0322	0.0800	[-0.1300, 0.1872]	No	[-0.0691, 0.1318]	No	0.012
Day of Year	0.0532	0.0672	[-0.0758, 0.1908]	No	[-0.0304, 0.1378]	No	0.012
Latitude	0.0863	0.0823	[-0.0764, 0.2455]	No	[-0.0192, 0.1924]	No	0.012
Longitude Model							
Year	0.0252	0.0849	[-0.1447, 0.1924]	No	[-0.0834, 0.1313]	No	0.013
Day of Year	0.0522	0.0675	[-0.0745, 0.1891]	No	[-0.0332, 0.1387]	No	0.013
Longitude	-0.0630	0.0848	[-0.2309, 0.0987]	No	[-0.1738, 0.0456]	No	0.013
% Area Galled							
Latitude Model							
Year	0.0219	0.0728	[-0.1229, 0.1620]	No	[-0.0710, 0.1146]	No	0.001
Day of Year	0.0234	0.0649	[-0.0987, 0.1575]	No	[-0.0586, 0.1077]	No	0.001
Latitude	0.0381	0.0713	[-0.1030, 0.1770]	No	[-0.0533, 0.1296]	No	0.001
Longitude Model							
Year	0.0063	0.0771	[-0.1441, 0.1569]	No	[-0.0952, 0.1049]	No	0.001
Day of Year	0.0205	0.0628	[-0.0981, 0.1457]	No	[-0.0590, 0.1020]	No	0.001
Longitude	-0.0020	0.0749	[-0.1508, 0.1435]	No	[-0.0964, 0.0947]	No	0.001
% Area Mined							
Latitude Model							
Year	0.1766	0.0780	[0.0231, 0.3254]	Yes	[0.0750, 0.2762]	Yes	0.008
Day of Year	0.0147	0.0696	[-0.1160, 0.1567]	No	[-0.0724, 0.1043]	No	0.008
Latitude	0.1222	0.0775	[-0.0330, 0.2731]	No	[0.0232, 0.2195]	Yes	0.008
Longitude Model							
Year	0.1726	0.0847	[0.0035, 0.3351]	Yes	[0.0643, 0.2808]	Yes	0.006
Day of Year	0.0094	0.0693	[-0.1217, 0.1507]	No	[-0.0781, 0.0968]	No	0.006
Longitude	-0.1035	0.0812	[-0.2641, 0.0529]	No	[-0.2060, 0.0014]	No	0.006
Specialized Frequency							
Latitude Model							
Year	0.1090	0.1163	[-0.1284, 0.3313]	No	[-0.0372, 0.2539]	No	0.031
Day of Year	0.0381	0.1059	[-0.1729, 0.2441]	No	[-0.0955, 0.1728]	No	0.031
Latitude	0.1280	0.1114	[-0.0874, 0.3490]	No	[-0.0128, 0.2704]	No	0.031
Longitude Model							
Year	0.0983	0.1243	[-0.1485, 0.3410]	No	[-0.0574, 0.2581]	No	0.031
Day of Year	0.0354	0.1042	[-0.1637, 0.2447]	No	[-0.0992, 0.1682]	No	0.031
Longitude	-0.0823	0.1229	[-0.3228, 0.1618]	No	[-0.2398, 0.0758]	No	0.031
Specialized Richness							
Latitude Model							
Year	0.0447	0.1500	[-0.2481, 0.3364]	No	[-0.1493, 0.2307]	No	0.189
Day of Year	0.0999	0.1326	[-0.1538, 0.3657]	No	[-0.0660, 0.2712]	No	0.189
Latitude	0.1412	0.1390	[-0.1349, 0.4146]	No	[-0.0363, 0.3142]	No	0.189
Longitude Model							
Year	0.0411	0.1546	[-0.2716, 0.3395]	No	[-0.1587, 0.2317]	No	0.189
Day of Year	0.0945	0.1366	[-0.1690, 0.3725]	No	[-0.0750, 0.2694]	No	0.189
Longitude	-0.0932	0.1468	[-0.3913, 0.1896]	No	[-0.2788, 0.0915]	No	0.189
All Frequency							
Latitude Model							
Year	0.0877	0.1164	[-0.1512, 0.3101]	No	[-0.0611, 0.2327]	No	0.035
Day of Year	0.0820	0.1042	[-0.1265, 0.2836]	No	[-0.0517, 0.2151]	No	0.035
Latitude	0.0791	0.1129	[-0.1435, 0.2945]	No	[-0.0653, 0.2196]	No	0.035
Longitude Model							
Year	0.0766	0.1266	[-0.1751, 0.3212]	No	[-0.0841, 0.2358]	No	0.035
Day of Year	0.0783	0.1072	[-0.1298, 0.2924]	No	[-0.0569, 0.2155]	No	0.035
Longitude	-0.0423	0.1245	[-0.2878, 0.2015]	No	[-0.2002, 0.1174]	No	0.035
All Richness							
Latitude Model							
Year	0.1005	0.1432	[-0.1862, 0.3872]	No	[-0.0821, 0.2778]	No	0.228
Day of Year	0.1590	0.1351	[-0.1031, 0.4327]	No	[-0.0113, 0.3311]	No	0.228
Latitude	0.1250	0.1322	[-0.1427, 0.3824]	No	[-0.0437, 0.2906]	No	0.228
Longitude Model							
Year	0.0710	0.1512	[-0.2315, 0.3682]	No	[-0.1196, 0.2634]	No	0.229
Day of Year	0.1517	0.1337	[-0.1035, 0.4201]	No	[-0.0155, 0.3203]	No	0.229
Longitude	-0.0486	0.1418	[-0.3342, 0.2245]	No	[-0.2280, 0.1302]	No	0.229

Table S3:

Time Model Outputs - Early-Season Leaves

Variable	Slope	SE	95% CI	Exclude 0?	80% CI	Exclude 0?	R^2
Chewing Frequency							
Year	0.0531	0.1062	[-0.1590, 0.2641]	No	[-0.0815, 0.1909]	No	0.027
Day of Year	-0.1059	0.1031	[-0.3058, 0.0932]	No	[-0.2417, 0.0270]	No	0.027
Year × Day of Year	-0.1504	0.1289	[-0.3983, 0.1013]	No	[-0.3170, 0.0136]	No	0.027
Gall Frequency							
Year	0.0188	0.0807	[-0.1371, 0.1780]	No	[-0.0843, 0.1236]	No	0.012
Day of Year	0.0105	0.0833	[-0.1500, 0.1716]	No	[-0.0982, 0.1166]	No	0.012
Year × Day of Year	0.0186	0.0991	[-0.1782, 0.2162]	No	[-0.1062, 0.1445]	No	0.012
Gall Richness							
Year	0.3318	0.7217	[-1.1509, 1.7604]	No	[-0.5459, 1.2128]	No	0.151
Day of Year	0.0573	0.6274	[-1.2015, 1.2690]	No	[-0.7419, 0.8459]	No	0.151
Year × Day of Year	0.2238	0.7733	[-1.3487, 1.7353]	No	[-0.7543, 1.1783]	No	0.151
Generalized Frequency							
Year	0.0093	0.0859	[-0.1591, 0.1794]	No	[-0.0999, 0.1201]	No	0.013
Day of Year	-0.0125	0.0834	[-0.1828, 0.1494]	No	[-0.1163, 0.0951]	No	0.013
Year × Day of Year	0.0539	0.1060	[-0.1593, 0.2604]	No	[-0.0800, 0.1900]	No	0.013
Generalized Richness							
Year	0.0673	0.2641	[-0.4383, 0.6063]	No	[-0.2666, 0.4081]	No	0.050
Day of Year	-0.0581	0.2654	[-0.5894, 0.4506]	No	[-0.4010, 0.2794]	No	0.050
Year × Day of Year	0.0938	0.3267	[-0.5487, 0.7390]	No	[-0.3303, 0.5136]	No	0.050
Mine Frequency							
Year	0.0254	0.0920	[-0.1525, 0.2107]	No	[-0.0922, 0.1458]	No	0.014
Day of Year	0.0402	0.0935	[-0.1484, 0.2243]	No	[-0.0807, 0.1583]	No	0.014
Year × Day of Year	0.0026	0.1145	[-0.2220, 0.2274]	No	[-0.1427, 0.1458]	No	0.014
Mine Richness							
Year	0.1758	0.2931	[-0.3672, 0.7667]	No	[-0.1862, 0.5593]	No	0.061
Day of Year	0.2071	0.2779	[-0.3454, 0.7526]	No	[-0.1510, 0.5557]	No	0.061
Year × Day of Year	-0.0715	0.3418	[-0.7504, 0.5966]	No	[-0.5068, 0.3558]	No	0.061
% Area Chewed							
Year	0.0039	0.0730	[-0.1394, 0.1494]	No	[-0.0881, 0.0981]	No	0.004
Day of Year	-0.0693	0.0742	[-0.2157, 0.0748]	No	[-0.1641, 0.0263]	No	0.004
Year × Day of Year	-0.0888	0.0917	[-0.2723, 0.0893]	No	[-0.2060, 0.0269]	No	0.004
% Area Galled							
Year	0.0222	0.0598	[-0.0928, 0.1419]	No	[-0.0537, 0.0994]	No	0.005
Day of Year	0.0562	0.0603	[-0.0639, 0.1738]	No	[-0.0217, 0.1331]	No	0.005
Year × Day of Year	0.0468	0.0736	[-0.0995, 0.1921]	No	[-0.0484, 0.1417]	No	0.005
% Area Mined							
Year	0.0343	0.0729	[-0.1078, 0.1797]	No	[-0.0577, 0.1287]	No	0.001
Day of Year	0.0171	0.0710	[-0.1212, 0.1558]	No	[-0.0737, 0.1081]	No	0.001
Year × Day of Year	0.0221	0.0864	[-0.1447, 0.1888]	No	[-0.0896, 0.1328]	No	0.001
Specialized Frequency							
Year	0.0912	0.1085	[-0.1290, 0.3005]	No	[-0.0463, 0.2286]	No	0.027
Day of Year	-0.0221	0.1077	[-0.2292, 0.1877]	No	[-0.1589, 0.1140]	No	0.027
Year × Day of Year	-0.1264	0.1311	[-0.3889, 0.1302]	No	[-0.2938, 0.0372]	No	0.027
Specialized Richness							
Year	0.1925	0.2366	[-0.2761, 0.6719]	No	[-0.1021, 0.4898]	No	0.301
Day of Year	-0.0050	0.2252	[-0.4440, 0.4526]	No	[-0.2859, 0.2777]	No	0.301
Year × Day of Year	-0.3283	0.2738	[-0.8760, 0.2116]	No	[-0.6742, 0.0157]	No	0.301
All Frequency							
Year	0.0980	0.1071	[-0.1097, 0.3070]	No	[-0.0367, 0.2357]	No	0.024
Day of Year	-0.0288	0.1068	[-0.2373, 0.1786]	No	[-0.1650, 0.1080]	No	0.024
Year × Day of Year	-0.0744	0.1296	[-0.3262, 0.1794]	No	[-0.2395, 0.0917]	No	0.024
All Richness							
Year	0.1410	0.2114	[-0.2804, 0.5534]	No	[-0.1232, 0.4078]	No	0.275
Day of Year	-0.0264	0.2039	[-0.4193, 0.3851]	No	[-0.2783, 0.2318]	No	0.275
Year × Day of Year	-0.2414	0.2479	[-0.7363, 0.2353]	No	[-0.5549, 0.0677]	No	0.275

Table S4:

Time Model Outputs - Late-Season Leaves

Variable	Slope	SE	95% CI	Exclude 0?	80% CI	Exclude 0?	R^2
Chewing Frequency							
Year	-0.0336	0.1089	[-0.2499, 0.1833]	No	[-0.1736, 0.1054]	No	0.027
Day of Year	0.0861	0.1066	[-0.1261, 0.2975]	No	[-0.0475, 0.2230]	No	0.027
Year × Day of Year	0.1293	0.1270	[-0.1222, 0.3836]	No	[-0.0308, 0.2893]	No	0.027
Gall Frequency							
Year	0.0697	0.0925	[-0.1109, 0.2561]	No	[-0.0480, 0.1861]	No	0.017
Day of Year	0.0597	0.0903	[-0.1170, 0.2394]	No	[-0.0553, 0.1765]	No	0.017
Year × Day of Year	0.0739	0.1110	[-0.1428, 0.2920]	No	[-0.0650, 0.2143]	No	0.017
Gall Richness							
Year	0.0601	0.2680	[-0.4846, 0.5741]	No	[-0.2768, 0.3953]	No	0.096
Day of Year	0.4376	0.3275	[-0.1681, 1.1274]	No	[0.0326, 0.8687]	Yes	0.096
Year × Day of Year	0.2756	0.3444	[-0.3879, 0.9754]	No	[-0.1500, 0.7088]	No	0.096
Generalized Frequency							
Year	-0.0078	0.0873	[-0.1773, 0.1598]	No	[-0.1202, 0.1040]	No	0.011
Day of Year	0.0515	0.0864	[-0.1130, 0.2279]	No	[-0.0574, 0.1631]	No	0.011
Year × Day of Year	0.0311	0.1015	[-0.1722, 0.2306]	No	[-0.0989, 0.1607]	No	0.011
Generalized Richness							
Year	0.2368	0.2233	[-0.2073, 0.6812]	No	[-0.0443, 0.5157]	No	0.151
Day of Year	0.5247	0.2983	[-0.0021, 1.1721]	No	[0.1623, 0.9097]	Yes	0.151
Year × Day of Year	-0.0012	0.3412	[-0.6975, 0.6609]	No	[-0.4310, 0.4257]	No	0.151
Mine Frequency							
Year	0.0194	0.0991	[-0.1766, 0.2123]	No	[-0.1068, 0.1459]	No	0.014
Day of Year	0.0326	0.0970	[-0.1541, 0.2252]	No	[-0.0912, 0.1581]	No	0.014
Year × Day of Year	0.0182	0.1163	[-0.2118, 0.2466]	No	[-0.1286, 0.1660]	No	0.014
Mine Richness							
Year	-0.0865	0.1918	[-0.4787, 0.2777]	No	[-0.3290, 0.1509]	No	0.089
Day of Year	0.1290	0.2063	[-0.2583, 0.5572]	No	[-0.1282, 0.3930]	No	0.089
Year × Day of Year	0.0155	0.2480	[-0.4975, 0.4781]	No	[-0.3008, 0.3200]	No	0.089
% Area Chewed							
Year	-0.0228	0.0739	[-0.1683, 0.1203]	No	[-0.1171, 0.0699]	No	0.009
Day of Year	0.0637	0.0705	[-0.0702, 0.2075]	No	[-0.0250, 0.1539]	No	0.009
Year × Day of Year	0.0533	0.0845	[-0.1144, 0.2194]	No	[-0.0536, 0.1576]	No	0.009
% Area Galled							
Year	0.0006	0.0676	[-0.1352, 0.1325]	No	[-0.0857, 0.0860]	No	0.001
Day of Year	0.0310	0.0661	[-0.0934, 0.1635]	No	[-0.0535, 0.1170]	No	0.001
Year × Day of Year	0.0183	0.0786	[-0.1349, 0.1698]	No	[-0.0822, 0.1194]	No	0.001
% Area Mined							
Year	0.1082	0.0722	[-0.0340, 0.2484]	No	[0.0149, 0.2010]	Yes	0.003
Day of Year	0.0119	0.0699	[-0.1197, 0.1553]	No	[-0.0759, 0.1035]	No	0.003
Year × Day of Year	0.0057	0.0862	[-0.1659, 0.1723]	No	[-0.1063, 0.1157]	No	0.003
Specialized Frequency							
Year	0.0180	0.1095	[-0.1994, 0.2324]	No	[-0.1201, 0.1562]	No	0.029
Day of Year	0.0706	0.1053	[-0.1300, 0.2780]	No	[-0.0636, 0.2041]	No	0.029
Year × Day of Year	0.1727	0.1272	[-0.0737, 0.4240]	No	[0.0113, 0.3322]	Yes	0.029
Specialized Richness							
Year	-0.0505	0.1461	[-0.3458, 0.2308]	No	[-0.2370, 0.1311]	No	0.232
Day of Year	0.1610	0.1530	[-0.1260, 0.4748]	No	[-0.0304, 0.3568]	No	0.232
Year × Day of Year	0.2022	0.1702	[-0.1302, 0.5425]	No	[-0.0122, 0.4181]	No	0.232
All Frequency							
Year	0.0138	0.1131	[-0.2128, 0.2339]	No	[-0.1304, 0.1535]	No	0.035
Day of Year	0.1199	0.1093	[-0.0920, 0.3355]	No	[-0.0174, 0.2600]	No	0.035
Year × Day of Year	0.2082	0.1289	[-0.0436, 0.4651]	No	[0.0463, 0.3707]	Yes	0.035
All Richness							
Year	-0.0032	0.1401	[-0.2803, 0.2630]	No	[-0.1819, 0.1715]	No	0.287
Day of Year	0.2176	0.1484	[-0.0587, 0.5263]	No	[0.0322, 0.4071]	Yes	0.287
Year × Day of Year	0.2110	0.1624	[-0.0986, 0.5351]	No	[0.0061, 0.4198]	Yes	0.287

Table S5:

	Dataset	Climate Variable (correlated with year)	Pearson r	P-value	Significance
Late-Season Leaves					
cor...1	**Old Leaves**	**AHM**	**0.404**	**0.0097**	**
cor...2	**Old Leaves**	**MAP**	**-0.341**	**0.0313**	*
cor...3	**Old Leaves**	**EXT**	**0.323**	**0.0423**	*
cor...4	Old Leaves	MAR	0.250	0.1194	ns
cor...5	Old Leaves	MSP	-0.196	0.2250	ns
cor...6	Old Leaves	SHM	0.177	0.2752	ns
cor...7	Old Leaves	RH	0.171	0.2908	ns
cor...8	Old Leaves	TD	0.076	0.6426	ns
Early-Season Leaves					
cor...9	**Young Leaves**	**EXT**	**0.350**	**0.0232**	*
cor...10	Young Leaves	MAR	0.203	0.1969	ns
cor...11	Young Leaves	SHM	-0.140	0.3774	ns
cor...12	Young Leaves	RH	-0.106	0.5050	ns
cor...13	Young Leaves	MSP	-0.062	0.6971	ns
cor...14	Young Leaves	AHM	0.001	0.9974	ns

Table S6:

Model outputs - Early-Season Leaves

Variable	Slope	SE	90% CI	Excl. 0?	80% CI	Excl. 0?
Cleaning Frequency						
AVM						
AVM	-0.1964	0.1045	[-0.4642, 0.0072]	No	[-0.3286, -0.0620]	Yes
Year	0.0672	0.1073	[-0.1429, 0.2810]	No	[-0.0701, 0.2024]	No
Year^2	-0.1389	0.1055	[-0.3448, 0.0671]	No	[-0.2775, -0.0011]	Yes
CST						
CST	-0.1122	0.1177	[-0.3403, 0.1226]	No	[-0.2608, 0.0367]	No
Year	0.1015	0.1114	[-0.1148, 0.3196]	No	[-0.0416, 0.2447]	No
Year^2	-0.1213	0.1062	[-0.3284, 0.0895]	No	[-0.2574, 0.0154]	No
MAX						
MAX	-0.2115	0.1063	[-0.4196, -0.0005]	Yes	[-0.3495, -0.0768]	Yes
Year	0.1149	0.1072	[-0.0926, 0.3287]	No	[-0.0211, 0.2494]	No
Year^2	-0.1549	0.1032	[-0.3577, 0.0468]	No	[-0.2379, -0.0279]	Yes
MDP						
MDP	0.0143	0.1090	[-0.1986, 0.2285]	No	[-0.1215, 0.1527]	No
Year	0.0835	0.1053	[-0.1418, 0.2741]	No	[-0.0708, 0.1963]	No
Year^2	-0.1027	0.1066	[-0.3123, 0.1055]	No	[-0.2262, 0.0324]	No
PAS						
PAS	-0.0885	0.1138	[-0.2405, 0.2296]	No	[-0.1611, 0.1414]	No
Year	0.0607	0.1120	[-0.1626, 0.2796]	No	[-0.0834, 0.2061]	No
Year^2	-0.0988	0.1140	[-0.3236, 0.1252]	No	[-0.2420, 0.0348]	No
Gall Frequency						
AVM						
AVM	-0.0243	0.0816	[-0.1856, 0.1356]	No	[-0.1276, 0.0788]	No
Year	0.0144	0.0865	[-0.1405, 0.1714]	No	[-0.0877, 0.1196]	No
Year^2	0.0841	0.0814	[-0.1514, 0.1905]	No	[-0.0973, 0.1746]	No
CST						
CST	-0.0282	0.0880	[-0.1996, 0.1430]	No	[-0.1407, 0.0837]	No
Year	0.0271	0.0866	[-0.1415, 0.2042]	No	[-0.0878, 0.1392]	No
Year^2	0.0952	0.0838	[-0.1585, 0.1685]	No	[-0.1028, 0.1139]	No
MAX						
MAX	-0.0322	0.0843	[-0.1966, 0.1425]	No	[-0.1304, 0.0812]	No
Year	0.0230	0.0852	[-0.1364, 0.1855]	No	[-0.0816, 0.1291]	No
Year^2	0.0107	0.0825	[-0.1539, 0.1771]	No	[-0.0946, 0.1148]	No
MDP						
MDP	-0.0146	0.0825	[-0.1779, 0.1417]	No	[-0.1223, 0.0919]	No
Year	0.0176	0.0815	[-0.1453, 0.1793]	No	[-0.0855, 0.1198]	No
Year^2	0.0119	0.0827	[-0.1501, 0.1756]	No	[-0.0938, 0.1161]	No
SHW						
SHW	-0.0137	0.0836	[-0.1917, 0.1465]	No	[-0.1275, 0.0867]	No
Year	0.0139	0.0817	[-0.1435, 0.1743]	No	[-0.0899, 0.1170]	No
Year^2	0.0936	0.0822	[-0.1485, 0.1723]	No	[-0.0992, 0.1170]	No
Gall Richness						
AVM						
AVM	-0.5420	0.7570	[-1.8558, 0.8513]	No	[-1.4004, 0.3261]	No
Year	0.3194	0.7479	[-1.2257, 1.8474]	No	[-0.5795, 1.2328]	No
Year^2	0.3623	0.6053	[-1.1148, 1.2845]	No	[-0.6965, 0.8278]	No
CST						
CST	-0.5369	0.6993	[-1.9197, 0.8738]	No	[-1.3876, 0.3305]	No
Year	0.4204	0.7420	[-1.1101, 1.8718]	No	[-0.4897, 1.3415]	No
Year^2	0.0913	0.5999	[-1.0915, 1.2704]	No	[-0.6635, 0.8479]	No
MAX						
MAX	-0.2457	0.6670	[-1.5395, 1.1763]	No	[-1.0717, 0.5628]	No
Year	0.5380	0.7334	[-1.2138, 1.7844]	No	[-0.5376, 1.2439]	No
Year^2	0.0449	0.6126	[-1.2027, 1.3549]	No	[-0.7746, 0.8014]	No
MDP						
MDP	-0.2254	0.7557	[-1.7129, 1.3513]	No	[-1.1696, 0.6986]	No
Year	0.3415	0.7459	[-1.1663, 1.8332]	No	[-0.5699, 1.2542]	No
Year^2	0.0730	0.6236	[-1.1968, 1.3443]	No	[-0.7175, 0.8431]	No
SHW						
SHW	-0.4059	0.8074	[-1.9809, 1.2626]	No	[-1.4123, 0.5007]	No
Year	0.2795	0.7737	[-1.2146, 1.7695]	No	[-0.5756, 1.1597]	No
Year^2	0.0965	0.7064	[-1.0630, 1.2690]	No	[-0.6416, 0.8472]	No
Generalized Frequency						
AVM						
AVM	-0.0125	0.0868	[-0.1804, 0.1569]	No	[-0.1228, 0.0965]	No
Year	0.0052	0.0858	[-0.1633, 0.1754]	No	[-0.1037, 0.1146]	No
Year^2	-0.0148	0.0861	[-0.1851, 0.1524]	No	[-0.1248, 0.0953]	No
CST						
CST	-0.0002	0.0935	[-0.1836, 0.1850]	No	[-0.1186, 0.1168]	No
Year	0.0041	0.0855	[-0.1756, 0.1900]	No	[-0.1113, 0.1241]	No
Year^2	-0.0128	0.0870	[-0.1838, 0.1092]	No	[-0.1252, 0.0996]	No
MAX						
MAX	-0.0095	0.0878	[-0.1797, 0.1648]	No	[-0.1228, 0.1028]	No
Year	0.0075	0.0888	[-0.1690, 0.1855]	No	[-0.1004, 0.1223]	No
Year^2	-0.0134	0.0865	[-0.1836, 0.1589]	No	[-0.1249, 0.0980]	No
MDP						
MDP	-0.0193	0.0848	[-0.1873, 0.1464]	No	[-0.1295, 0.0988]	No
Year	0.0047	0.0871	[-0.1661, 0.1793]	No	[-0.1077, 0.1172]	No
Year^2	-0.0105	0.0859	[-0.1774, 0.1601]	No	[-0.1196, 0.0991]	No
SHW						
SHW	-0.0207	0.0874	[-0.1910, 0.1604]	No	[-0.1328, 0.0894]	No
Year	0.0034	0.0896	[-0.1693, 0.1796]	No	[-0.1123, 0.1180]	No
Year^2	-0.0114	0.0857	[-0.1796, 0.1561]	No	[-0.1235, 0.0998]	No
Generalized Richness						
AVM						
AVM	-0.4496	0.3016	[-1.0114, 0.1025]	No	[-0.8387, 0.0748]	No
Year	0.0963	0.2910	[-0.4427, 0.6441]	No	[-0.2791, 0.4679]	No
Year^2	-0.1749	0.2728	[-0.6364, 0.3959]	No	[-0.4691, 0.2203]	No
CST						
CST	-0.1777	0.3952	[-0.7453, 0.3566]	No	[-0.5653, 0.1968]	No
Year	0.1175	0.2798	[-0.4138, 0.6767]	No	[-0.2391, 0.4773]	No
Year^2	-0.0569	0.2724	[-0.6021, 0.4620]	No	[-0.4073, 0.2981]	No
MAX						
MAX	-0.2218	0.2753	[-0.7595, 0.2970]	No	[-0.5676, 0.1157]	No
Year	0.0962	0.3014	[-0.4053, 0.6177]	No	[-0.2296, 0.4321]	No
Year^2	-0.0671	0.2635	[-0.5365, 0.4398]	No	[-0.4021, 0.2597]	No
MDP						
MDP	0.0322	0.2545	[-0.4817, 0.5138]	No	[-0.3093, 0.3481]	No
Year	0.0681	0.2910	[-0.4263, 0.5657]	No	[-0.2646, 0.3968]	No
Year^2	-0.0452	0.2719	[-0.5071, 0.4802]	No	[-0.3914, 0.2965]	No
SHW						
SHW	-0.0919	0.2543	[-0.5551, 0.4564]	No	[-0.3269, 0.3964]	No
Year	0.0585	0.2660	[-0.4129, 0.5671]	No	[-0.2647, 0.3949]	No
Year^2	0.0564	0.2640	[-0.3914, 0.5441]	No	[-0.3016, 0.3814]	No
Mine Frequency						
AVM						
AVM	-0.0328	0.0914	[-0.2143, 0.1440]	No	[-0.1487, 0.0840]	No
Year	0.0245	0.0908	[-0.1506, 0.2050]	No	[-0.0913, 0.1411]	No
Year^2	0.0006	0.0932	[-0.1516, 0.2120]	No	[-0.0857, 0.1010]	No
CST						
CST	-0.0187	0.1021	[-0.2229, 0.1779]	No	[-0.1510, 0.1097]	No
Year	0.0315	0.0977	[-0.1628, 0.2246]	No	[-0.0962, 0.1542]	No
Year^2	0.0082	0.0909	[-0.1397, 0.2177]	No	[-0.0785, 0.1032]	No
MAX						
MAX	-0.1031	0.0917	[-0.2857, 0.0793]	No	[-0.2211, 0.0158]	No
Year	0.0498	0.0921	[-0.1363, 0.2350]	No	[-0.0862, 0.1427]	No
Year^2	0.0425	0.0925	[-0.1408, 0.2249]	No	[-0.0779, 0.1704]	No
MDP						
MDP	0.0407	0.0935	[-0.1434, 0.2248]	No	[-0.0782, 0.1630]	No
Year	0.0275	0.0902	[-0.1504, 0.2063]	No	[-0.0986, 0.1423]	No
Year^2	0.0367	0.0917	[-0.1426, 0.2142]	No	[-0.0815, 0.1545]	No
SHW						
SHW	-0.0103	0.0928	[-0.1945, 0.1693]	No	[-0.1301, 0.1087]	No
Year	0.0338	0.0927	[-0.1375, 0.2083]	No	[-0.0937, 0.1421]	No
Year^2	0.0407	0.0926	[-0.1425, 0.2279]	No	[-0.0791, 0.1785]	No
Mine Richness						
AVM						
AVM	-0.2448	0.2114	[-0.6962, 0.2306]	No	[-0.5425, 0.1448]	No
Year	0.1356	0.2650	[-0.3991, 0.7622]	No	[-0.2079, 0.5289]	No
Year^2	0.1764	0.2720	[-0.3614, 0.7762]	No	[-0.1774, 0.5327]	No
CST						
CST	-0.1490	0.3065	[-0.7543, 0.4456]	No	[-0.5365, 0.2371]	No
Year	0.2222	0.3200	[-0.3843, 0.8858]	No	[-0.1766, 0.6370]	No
Year^2	0.1904	0.2675	[-0.3411, 0.7964]	No	[-0.1479, 0.5327]	No
MAX						
MAX	-0.6280	0.3210	[-1.2912, -0.0311]	Yes	[-1.0414, -0.2200]	Yes
Year	0.2899	0.2963	[-0.2804, 0.8803]	No	[-0.0796, 0.6681]	No
Year^2	0.1521	0.2795	[-0.3957, 0.6848]	No	[-0.1986, 0.5007]	No
MDP						
MDP	0.2038	0.2593	[-0.3113, 0.7542]	No	[-0.1228, 0.5314]	No
Year	0.1982	0.2874	[-0.3350, 0.7549]	No	[-0.1545, 0.5649]	No
Year^2	0.1969	0.2731	[-0.3684, 0.7253]	No	[-0.1427, 0.5312]	No
SHW						
SHW	-0.0922	0.2184	[-0.7679, 0.4875]	No	[-0.5049, 0.2847]	No
Year	0.1562	0.2979	[-0.4012, 0.7705]	No	[-0.2141, 0.5393]	No
Year^2	0.1914	0.2791	[-0.3571, 0.7372]	No	[-0.1549, 0.5422]	No
% Area Closed						
AVM						
AVM	-0.1011	0.0749	[-0.2490, 0.0451]	No	[-0.1981, -0.0042]	Yes
Year	0.0138	0.0719	[-0.1257, 0.1055]	No	[-0.0781, 0.1059]	No
Year^2	-0.0867	0.0775	[-0.2403, 0.0597]	No	[-0.1872, 0.0058]	No
CST						
CST	-0.0601	0.0823	[-0.2247, 0.0999]	No	[-0.1648, 0.0431]	No
Year	0.0325	0.0795	[-0.1302, 0.1965]	No	[-0.0871, 0.1442]	No
Year^2	-0.0797	0.0763	[-0.2275, 0.0653]	No	[-0.1775, 0.0180]	No
MAX						
MAX	-0.0875	0.0746	[-0.2346, 0.0562]	No	[-0.1827, 0.0063]	No
Year	0.0336	0.0744	[-0.1169, 0.1796]	No	[-0.0590, 0.1241]	No
Year^2	-0.0736	0.0765	[-0.2262, 0.0714]	No	[-0.1707, 0.0237]	No
MDP						
MDP	0.0160	0.0844	[-0.1563, 0.1925]	No	[-0.0955, 0.1257]	No
Year	0.0191	0.0777	[-0.1537, 0.1697]	No	[-0.0885, 0.1182]	No
Year^2	-0.0768	0.0812	[-0.2364, 0.0813]	No	[-0.1786, 0.0255]	No
SHW						
SHW	0.0871	0.0762	[-0.0708, 0.2342]	No	[-0.0114, 0.1904]	No
Year	0.0222	0.0741	[-0.1181, 0.1696]	No	[-0.0704, 0.1719]	No
Year^2	-0.0723	0.0747	[-0.2207, 0.0719]	No	[-0.1681, 0.0228]	No
% Area Gulled						
AVM						
AVM	0.0965	0.0501	[-0.1080, 0.1177]	No	[-0.0683, 0.2803]	No
Year	0.0196	0.0593	[-0.0963, 0.1371]	No	[-0.0566, 0.0960]	No
Year^2	0.0965	0.0614	[-0.0849, 0.1706]	No	[-0.0325, 0.1965]	No
CST						
CST	0.0261	0.0648	[-0.1033, 0.1549]	No	[-0.0555, 0.1164]	No
Year	0.0113	0.0626	[-0.1157, 0.1414]	No	[-0.0691, 0.0941]	No
Year^2	0.0608	0.0616	[-0.0595, 0.1902]	No	[-0.0197, 0.1358]	No
MAX						
MAX	-0.0152	0.0596	[-0.1347, 0.0946]	No	[-0.0925, 0.0601]	No
Year	0.0245	0.0613	[-0.0927, 0.1472]	No	[-0.0535, 0.1049]	No
Year^2	0.0544	0.0595	[-0.0647, 0.1693]	No	[-0.0325, 0.1313]	No
RH						
RH	-0.0010	0.0588	[-0.1173, 0.1146]	No	[-0.0752, 0.0741]	No
Year	0.0211	0.0596	[-0.0966, 0.1384]	No	[-0.0541, 0.0973]	No
Year^2	0.0551	0.0598	[-0.0626, 0.1733]	No	[-0.0224, 0.1387]	No
SHW						
SHW	-0.0259	0.0618	[-0.1542, 0.0874]	No	[-0.1057, 0.0543]	No
Year	0.0174	0.0601	[-0.0993, 0.1362]	No	[-0.0590, 0.0941]	No
Year^2	0.0564	0.0605	[-0.0591, 0.1771]	No	[-0.0206, 0.1347]	No
% Area Killed						
AVM						
AVM	-0.0205	0.0703	[-0.1612, 0.1164]	No	[-0.1157, 0.0750]	No
Year	0.0313	0.0713	[-0.1038, 0.1761]	No	[-0.0585, 0.1236]	No
Year^2	0.0135	0.0703	[-0.1328, 0.1595]	No	[-0.0761, 0.1029]	No
CST						
CST	-0.0196	0.0754	[-0.1703, 0.1265]	No	[-0.1143, 0.0750]	No
Year	0.0366	0.0756	[-0.1050, 0.1980]	No	[-0.0579, 0.1307]	No
Year^2	0.0157	0.0722	[-0.1264, 0.1708]	No	[-0.0776, 0.1077]	No
MAX						
MAX	-0.0213	0.0697	[-0.1591, 0.1158]	No	[-0.1105, 0.0673]	No
Year	0.0376	0.0718	[-0.1019, 0.1795]	No	[-0.0541, 0.1299]	No
Year^2	0.0164	0.0705	[-0.1292, 0.1540]	No	[-0.0742, 0.1050]	No
MDP						
MDP	0.0004	0.0664	[-0.1309, 0.1329]	No	[-0.0882, 0.0877]	No
Year	0.0341	0.0712	[-0.1029, 0.1798]	No	[-0.0562, 0.1251]	No
Year^2	0.0178	0.0696	[-0.1179, 0.1521]	No	[-0.0771, 0.1085]	No
SHW						
SHW	-0.0263	0.0720	[-0.1707, 0.1063]	No	[-0.1195, 0.0629]	No
Year	0.0295	0.0701	[-0.1079, 0.1641]	No	[-0.0599, 0.1321]	No
Year^2	0.0164	0.0706	[-0.1214, 0.1545]	No	[-0.0764, 0.1068]	No
Specialized Frequency						
AVM						
AVM	-0.1963	0.1029	[-0.3991, 0.0104]	No	[-0.3299, -0.0643]	Yes
Year	0.1034	0.1040	[-0.0996, 0.3066]	No	[-0.0276, 0.2257]	No
Year^2	-0.0016	0.1079	[-0.2628, 0.1632]	No	[-0.1884, 0.0852]	No
CST						
CST	-0.1024	0.1114	[-0.3239, 0.1135]	No	[-0.2521, 0.0410]	No
Year	0.1161	0.1148	[-0.0897, 0.3353]	No	[-0.0319, 0.2795]	No
Year^2	-0.0358	0.1095	[-0.2445, 0.1902]	No	[-0.1745, 0.1020]	No
MAX						
MAX	-0.2375	0.1054	[-0.4405, -0.0247]	Yes	[-0.3792, -0.1009]	Yes
Year	0.1571	0.1095	[-0.0525, 0.3730]	No	[0.0199, 0.2963]	Yes
Year^2	-0.0181	0.1054	[-0.2203, 0.1866]	No	[-0.1550, 0.1154]	No
MDP						
MDP	0.0403	0.1086	[-0.1737, 0.2445]	No	[-0.0896, 0.1780]	No
Year	0.1024	0.1109	[-0.1066, 0.3121]	No	[-0.0315, 0.2389]	No
Year^2	-0.0187	0.1080	[-0.2266, 0.1949]	No	[-0.1355, 0.1186]	No
PAS						
PAS	-0.0710	0.1225	[-0.3125, 0.2283]	No	[-0.1675, 0.1467]	No
Year	0.0963	0.1122	[-0.1118, 0.3225]	No	[-0.0447, 0.2414]	No
Year^2	-0.0147	0.1161	[-0.2429, 0.2135]	No	[-0.1663, 0.1353]	No
Specialized Richness						
AVM						
AVM	-0.4050	0.2483	[-0.9863, 0.0032]	No	[-0.7690, -0.1398]	Yes
Year	0.1600	0.2532	[-0.2951, 0.6204]	No	[-0.1195, 0.4578]	No
Year^2	-0.0827	0.2331	[-0.4877, 0.3859]	No	[-0.3306, 0.2376]	No
CST						
CST	-0.3297	0.2997	[-0.8109, 0.1321]	No	[-0.6233, -0.0163]	Yes
Year	0.2427	0.2514	[-0.1562, 0.6464]	No	[0.0233, 0.4627]	Yes
Year^2	-0.0984	0.2330	[-0.4763, 0.2990]	No	[-0.3329, 0.1456]	No
MAX						
MAX	-0.0110	0.2531	[-0.1727, 0.7354]	No	[-0.1646, 0.2850]	No
Year	0.3555	0.2555	[0.1175, 0.8109]	Yes	[0.0429, 0.6524]	Yes
Year^2	-0.0589	0.2223	[-0.4891, 0.3801]	No	[-0.3336, 0.2152]	No
MDP						
MDP	-0.0196	0.2591	[-0.4542, 0.4147]	No	[-0.2931, 0.2697]	No
Year	0.2326	0.2564	[-0.2584, 0.7021]	No	[-0.0486, 0.5197]	No
Year^2	-0.0202	0.2278	[-0.4595, 0.4377]	No	[-0.3065, 0.2670]	No
PAS						
PAS	-0.1455	0.2538	[-0.6253, 0.3430]	No	[-0.4699, 0.1754]	No
Year	0.1898	0.2456	[0.2967, 0.6810]	Yes	[0.1166, 0.5507]	Yes
Year^2	0.0879	0.2323	[-0.3417, 0.4963]	No	[-0.2805, 0.2995]	No
All Frequency						
AVM						
AVM	-0.0096	0.1669	[-0.3179, 0.3050]	No	[-0.2439, 0.2256]	No
Year	0.1090	0.1586	[-0.0981, 0.3138]	No	[-0.0285, 0.2456]	No
Year^2	-0.0637	0.1673	[-0.2729, 0.3446]	No	[-0.1986, 0.0758]	No
CST						
CST	-0.1025	0.1766	[-0.3060, 0.1243]	No	[-0.2313, 0.0460]	No
Year	0.1372	0.1747	[-0.0841, 0.3584]	No	[-0.0119, 0.2825]	No
Year^2	-0.0419	0.1730	[-0.2604, 0.1687]	No	[-0.1987, 0.0924]	No
MAX						
MAX	-0.2459	0.1665	[-0.4552, -0.0390]	Yes	[-0.3830, -0.1113]	Yes
Year	0.1669	0.1670	[-0.0356, 0.3679]	No	[0.0236, 0.2985]	Yes
Year^2	-0.0290	0.1551	[-0.2285, 0.1783]	No	[-0.1929, 0.1380]	No
MDP						
MDP	0.0230	0.1732	[-0.1914, 0.2437]	No	[-0.1166, 0.1641]	No
Year	0.1560	0.1846	[-0.1071, 0.5213]	No	[-0.0323, 0.5413]	No
Year^2	-0.0350	0.1582	[-0.2412, 0.1832]	No	[-0.1972, 0.1988]	No
PAS						
PAS	-0.1409	0.2384	[-0.6281, 0.2950]	No	[-0.4595, 0.1780]	No
Year	0.1537	0.2208	[-0.3139, 0.6200]	No	[-0.1547, 0.4608]	No
Year^2	0.0897	0.2142	[-0.4132, 0.4256]	No	[-0.2756, 0.2905]	No

Table S7:

Model outputs - Late-Season Leaves

Variable	Slope	SE	90% CI	Excl. 0?	80% CI	Excl. 0?	R²
Cleaning Frequency							
CIT							
MAX	0.0175	0.1153	[-0.2091, 0.2422]	No	[-0.1281, 0.1548]	No	0.026
Year	-0.0132	0.1166	[-0.2043, 0.2395]	No	[-0.1023, 0.1867]	No	0.026
Day of Year	0.0627	0.1041	[-0.1405, 0.2686]	No	[-0.0871, 0.1948]	No	0.026
MAP							
MAX	0.0578	0.1096	[-0.1421, 0.2676]	No	[-0.0810, 0.1966]	No	0.026
Year	-0.0208	0.1137	[-0.2064, 0.1961]	No	[-0.1419, 0.1764]	No	0.026
Day of Year	0.0550	0.1049	[-0.1434, 0.2621]	No	[-0.0777, 0.1874]	No	0.026
MXP							
MAX	-0.1263	0.1086	[-0.3431, 0.0906]	No	[-0.2645, 0.0097]	No	0.028
Year	-0.0375	0.1112	[-0.2597, 0.1777]	No	[-0.1811, 0.1014]	No	0.028
Day of Year	0.0921	0.1067	[-0.1124, 0.2995]	No	[-0.0404, 0.2245]	No	0.028
Rh							
Rh	-0.1725	0.1028	[-0.3727, 0.0370]	No	[-0.3050, 0.0474]	No	0.029
Year	0.0185	0.1052	[-0.1885, 0.2208]	No	[-0.1148, 0.1508]	No	0.029
Day of Year	0.0790	0.1006	[-0.1187, 0.2764]	No	[-0.0484, 0.2079]	No	0.029
SHW							
MAX	-0.0640	0.1089	[-0.2756, 0.2141]	No	[-0.1411, 0.1327]	No	0.027
Year	-0.0064	0.1136	[-0.2268, 0.2360]	No	[-0.1461, 0.1540]	No	0.027
Day of Year	0.0640	0.1061	[-0.1480, 0.2723]	No	[-0.0736, 0.1963]	No	0.027
Gall Frequency							
CIT							
MAX	-0.0469	0.0957	[-0.1434, 0.2345]	No	[-0.0764, 0.1686]	No	0.017
Year	0.0671	0.0952	[-0.1194, 0.2543]	No	[-0.0538, 0.1955]	No	0.017
Day of Year	0.0047	0.0895	[-0.1533, 0.2372]	No	[-0.0715, 0.1766]	No	0.017
MAP							
MAX	-0.0701	0.0960	[-0.2595, 0.1141]	No	[-0.1945, 0.0534]	No	0.017
Year	0.0571	0.0954	[-0.1364, 0.2348]	No	[-0.0834, 0.1794]	No	0.017
Day of Year	0.0694	0.0936	[-0.1139, 0.2534]	No	[-0.0501, 0.1893]	No	0.017
MXP							
MAX	0.0209	0.0922	[-0.1522, 0.2095]	No	[-0.0878, 0.1475]	No	0.016
Year	0.0760	0.0914	[-0.1031, 0.2513]	No	[-0.0429, 0.1931]	No	0.016
Day of Year	0.0428	0.0878	[-0.1129, 0.2164]	No	[-0.0877, 0.1541]	No	0.016
Rh							
MXP	-0.1563	0.0896	[-0.3106, 0.0398]	No	[-0.2493, 0.0205]	No	0.022
Year	0.0571	0.0885	[-0.1177, 0.2286]	No	[-0.0555, 0.1699]	No	0.022
Day of Year	0.0806	0.0912	[-0.0995, 0.2620]	No	[-0.0242, 0.1960]	No	0.022
SHW							
MAX	-0.2093	0.0945	[0.0313, 0.3818]	Yes	[0.0936, 0.3245]	Yes	0.038
Year	0.0626	0.0961	[-0.1074, 0.2363]	No	[-0.0443, 0.1756]	No	0.038
Day of Year	0.0598	0.0859	[-0.1258, 0.2110]	No	[-0.0477, 0.1646]	No	0.038
Gall Richness							
CIT							
MAX	0.2745	0.2352	[-0.2898, 0.7836]	No	[-0.0498, 0.6077]	No	0.092
Year	0.0478	0.2443	[-0.4535, 0.5323]	No	[-0.2626, 0.3539]	No	0.092
Day of Year	0.2934	0.2306	[-0.2423, 0.9150]	No	[-0.0690, 0.6812]	No	0.092
MAP							
MAX	-0.4765	0.2336	[-1.1424, 0.1385]	No	[-0.9143, -0.0449]	Yes	0.087
Year	-0.0201	0.2556	[-0.7236, 0.4803]	No	[-0.3483, 0.2942]	No	0.087
Day of Year	0.4217	0.2503	[-0.1235, 1.0246]	No	[0.0554, 0.8070]	Yes	0.087
MXP							
MAX	0.1154	0.2420	[-0.3575, 0.6862]	No	[-0.1822, 0.4227]	No	0.094
Year	0.1342	0.2450	[-0.3035, 0.5957]	No	[-0.1836, 0.4387]	No	0.094
Day of Year	0.2944	0.2435	[-0.2565, 0.9404]	No	[-0.0864, 0.6870]	No	0.094
MXP							
MXP	-0.6995	0.2260	[-1.0875, -0.1526]	Yes	[-0.9065, -0.0721]	Yes	0.138
Year	0.0688	0.2179	[-0.2936, 0.5550]	No	[-0.2005, 0.3348]	No	0.138
Day of Year	0.3128	0.2579	[-0.1508, 0.8743]	No	[0.0011, 0.6474]	Yes	0.138
SHW							
MAX	0.4959	0.1606	[0.1942, 0.8208]	Yes	[0.2943, 0.6946]	Yes	0.137
Year	0.0960	0.2077	[-0.3279, 0.4958]	No	[-0.1575, 0.3501]	No	0.137
Day of Year	0.1374	0.2467	[-0.3057, 0.5603]	No	[-0.1773, 0.4483]	No	0.137
Generalized Frequency							
CIT							
MAX	-0.0607	0.0891	[-0.2188, 0.1314]	No	[-0.1974, 0.0609]	No	0.012
Year	0.0174	0.0884	[-0.1582, 0.1936]	No	[-0.0947, 0.1320]	No	0.012
Day of Year	0.0476	0.0855	[-0.1130, 0.2126]	No	[-0.0572, 0.1555]	No	0.012
MAP							
MAX	-0.1387	0.0893	[-0.0384, 0.3344]	No	[0.0124, 0.2650]	Yes	0.020
Year	0.0617	0.0952	[-0.1255, 0.2327]	No	[-0.0616, 0.1918]	No	0.020
Day of Year	0.0935	0.0868	[-0.1762, 0.2756]	No	[-0.1101, 0.1116]	No	0.020
MXP							
MAX	0.0947	0.0846	[-0.1653, 0.1689]	No	[-0.1346, 0.1129]	No	0.011
Year	-0.0923	0.0852	[-0.1795, 0.1072]	No	[-0.1105, 0.0340]	No	0.011
Day of Year	0.0414	0.0825	[-0.1186, 0.2005]	No	[-0.0837, 0.1467]	No	0.011
MXP							
MXP	0.0413	0.0850	[-0.1252, 0.2093]	No	[-0.0876, 0.1514]	No	0.012
Year	0.0108	0.0865	[-0.1587, 0.1793]	No	[-0.0997, 0.1214]	No	0.012
Day of Year	0.0339	0.0852	[-0.1314, 0.2060]	No	[-0.0732, 0.1424]	No	0.012
SHW							
MAX	-0.0209	0.0835	[-0.1605, 0.1410]	No	[-0.1281, 0.0862]	No	0.011
Year	0.0645	0.0850	[-0.1009, 0.1752]	No	[-0.1049, 0.1142]	No	0.011
Day of Year	0.0428	0.0813	[-0.1157, 0.2019]	No	[-0.0811, 0.1477]	No	0.011
Generalized Richness							
AHW							
MAX	-0.0278	0.2503	[-0.5371, 0.4457]	No	[-0.3428, 0.2844]	No	0.155
Year	0.2573	0.2334	[-0.1944, 0.7447]	No	[-0.0377, 0.5461]	No	0.155
Day of Year	0.4779	0.2077	[0.0850, 1.1414]	Yes	[0.1894, 0.8755]	Yes	0.155
MAX							
MAX	0.0424	0.2077	[-0.3605, 0.4595]	No	[-0.2206, 0.3043]	No	0.157
Year	0.2164	0.2579	[-0.3966, 0.8314]	No	[-0.0915, 0.8018]	No	0.157
Day of Year	0.4899	0.2921	[-0.0284, 1.1386]	No	[0.1364, 0.8435]	Yes	0.157
MXP							
MXP	-0.0563	0.2119	[-0.4705, 0.3701]	No	[-0.3142, 0.2177]	No	0.156
Year	0.2396	0.2033	[-0.1969, 0.6574]	No	[-0.0123, 0.4928]	No	0.156
Day of Year	0.4964	0.2930	[-0.0128, 1.1584]	No	[0.1429, 0.8750]	Yes	0.156
Rh							
Rh	-0.0645	0.1830	[-0.4331, 0.3071]	No	[-0.2941, 0.1630]	No	0.155
Year	0.2450	0.1964	[-0.1573, 0.6462]	No	[0.0005, 0.4941]	Yes	0.155
Day of Year	0.4916	0.2951	[-0.0397, 1.1311]	No	[0.1336, 0.8703]	Yes	0.155
SHW							
MAX	-0.0455	0.1941	[-0.4394, 0.3272]	No	[-0.2929, 0.1949]	No	0.158
Year	0.2400	0.2026	[-0.1542, 0.6556]	No	[-0.0887, 0.4993]	No	0.158
Day of Year	0.5106	0.3033	[-0.0320, 1.1756]	No	[0.1467, 0.8633]	Yes	0.158
Mine Frequency							
AHW							
MAX	-0.0425	0.1164	[-0.2687, 0.1857]	No	[-0.1925, 0.1033]	No	0.014
Year	0.0476	0.1119	[-0.1729, 0.2687]	No	[-0.0968, 0.1919]	No	0.014
Day of Year	0.0104	0.1051	[-0.1973, 0.2192]	No	[-0.1255, 0.1481]	No	0.014
CIT							
MAX	-0.1315	0.1048	[-0.3376, 0.0795]	No	[-0.2638, 0.0037]	No	0.018
Year	0.0773	0.1055	[-0.1336, 0.2804]	No	[-0.0558, 0.2114]	No	0.018
Day of Year	0.0376	0.0992	[-0.1389, 0.2142]	No	[-0.0754, 0.1520]	No	0.018
MAP							
MAX	0.0307	0.1042	[-0.1723, 0.2267]	No	[-0.1027, 0.1614]	No	0.014
Year	0.0343	0.1045	[-0.1713, 0.2383]	No	[-0.0990, 0.1669]	No	0.014
Day of Year	0.0208	0.0975	[-0.1481, 0.2134]	No	[-0.1051, 0.1448]	No	0.014
MXP							
MAX	-0.0477	0.0966	[-0.2413, 0.1386]	No	[-0.1609, 0.0746]	No	0.014
Year	0.0379	0.0994	[-0.1364, 0.2351]	No	[-0.0875, 0.1642]	No	0.014
Day of Year	0.0242	0.0901	[-0.1460, 0.2124]	No	[-0.0795, 0.1471]	No	0.014
SHW							
MAX	-0.0797	0.0911	[-0.2571, 0.1015]	No	[-0.1908, 0.0376]	No	0.015
Year	0.0564	0.0958	[-0.1521, 0.2233]	No	[-0.0864, 0.1587]	No	0.015
Day of Year	0.0561	0.0955	[-0.1472, 0.2270]	No	[-0.0860, 0.1588]	No	0.015
Mine Richness							
AHW							
MAX	-0.0601	0.2234	[-0.5399, 0.3966]	No	[-0.3479, 0.2227]	No	0.089
Year	-0.0533	0.2129	[-0.4879, 0.3524]	No	[-0.3236, 0.2109]	No	0.089
Day of Year	0.0863	0.2201	[-0.3383, 0.5278]	No	[-0.1909, 0.3632]	No	0.089
CIT							
MAX	-0.2189	0.1736	[-0.5734, 0.1449]	No	[-0.4466, 0.0356]	No	0.090
Year	0.0828	0.1848	[-0.3793, 0.5910]	No	[-0.2206, 0.3939]	No	0.090
Day of Year	0.1246	0.2003	[-0.2904, 0.5560]	No	[-0.1184, 0.3814]	No	0.090
MAP							
MAX	-0.0159	0.1986	[-0.3874, 0.3543]	No	[-0.2497, 0.2254]	No	0.088
Year	-0.0858	0.1934	[-0.4898, 0.2737]	No	[-0.3316, 0.1521]	No	0.088
Day of Year	0.1166	0.2076	[-0.2791, 0.5443]	No	[-0.1463, 0.3753]	No	0.088
MXP							
MAX	-0.1495	0.1917	[-0.5333, 0.2132]	No	[-0.3911, 0.0936]	No	0.091
Year	-0.0398	0.1875	[-0.4276, 0.3122]	No	[-0.2786, 0.1912]	No	0.091
Day of Year	0.1326	0.2041	[-0.2441, 0.5442]	No	[-0.1159, 0.3930]	No	0.091
SHW							
MAX	-0.1560	0.1950	[-0.5465, 0.2201]	No	[-0.4008, 0.0957]	No	0.095
Year	-0.0509	0.1790	[-0.4376, 0.2792]	No	[-0.2894, 0.1925]	No	0.095
Day of Year	0.1993	0.1980	[-0.2497, 0.6206]	No	[-0.1106, 0.5097]	No	0.095
Tx Area Closed							
CIT							
MAX	-0.0014	0.0799	[-0.1552, 0.1356]	No	[-0.1099, 0.0963]	No	0.009
Year	-0.0114	0.0768	[-0.1423, 0.1391]	No	[-0.1093, 0.0855]	No	0.009
Day of Year	0.0263	0.0755	[-0.0865, 0.1961]	No	[-0.0584, 0.1405]	No	0.009
MAP							
MAX	0.0465	0.0756	[-0.1026, 0.1952]	No	[-0.0480, 0.1418]	No	0.009
Year	-0.0266	0.0752	[-0.1782, 0.1158]	No	[-0.1225, 0.0873]	No	0.009
Day of Year	0.0452	0.0690	[-0.0853, 0.1962]	No	[-0.0495, 0.1320]	No	0.009
MXP							
MAX	-0.0750	0.0732	[-0.2215, 0.0649]	No	[-0.1665, 0.0174]	No	0.010
Year	-0.0275	0.0747	[-0.1769, 0.1192]	No	[-0.1207, 0.0644]	No	0.010
Day of Year	0.0633	0.0705	[-0.0844, 0.2128]	No	[-0.0276, 0.1575]	No	0.010
Rh							
Rh	-0.0902	0.0712	[-0.2374, 0.0446]	No	[-0.1816, 0.0002]	No	0.009
Year	0.0647	0.0705	[-0.1368, 0.1413]	No	[-0.0886, 0.0920]	No	0.009
Day of Year	0.0563	0.0659	[-0.0760, 0.1901]	No	[-0.0260, 0.1386]	No	0.009
SHW							
MAX	-0.0711	0.0720	[-0.1962, 0.1343]	No	[-0.1628, 0.0798]	No	0.009
Year	-0.0116	0.0714	[-0.1342, 0.2144]	No	[-0.1025, 0.0803]	No	0.009
Day of Year	0.0503	0.0675	[-0.0734, 0.1991]	No	[-0.0326, 0.1330]	No	0.009
Tx Area Edited							
CIT							
MAX	0.0149	0.0711	[-0.1235, 0.1540]	No	[-0.0796, 0.1368]	No	0.001
Year	-0.0011	0.0731	[-0.1460, 0.1436]	No	[-0.0972, 0.0914]	No	0.001
Day of Year	0.0193	0.0628	[-0.1013, 0.1457]	No	[-0.0595, 0.1011]	No	0.001
MAP							
MAX	-0.0162	0.0737	[-0.1663, 0.1201]	No	[-0.1124, 0.0768]	No	0.001
Year	0.0068	0.0713	[-0.1399, 0.1402]	No	[-0.0889, 0.0825]	No	0.001
Day of Year	0.0254	0.0684	[-0.1032, 0.1622]	No	[-0.0597, 0.1147]	No	0.001
MXP							
MAX	0.0274	0.0652	[-0.1055, 0.1564]	No	[-0.0557, 0.1096]	No	0.001
Year	-0.0019	0.0679	[-0.1358, 0.1371]	No	[-0.0896, 0.0849]	No	0.001
Day of Year	0.0163	0.0622	[-0.1019, 0.1419]	No	[-0.0616, 0.0905]	No	0.001
SHW							
MAX	0.0093	0.0637	[-0.1172, 0.1346]	No	[-0.0732, 0.0906]	No	0.001
Year	0.0007	0.0663	[-0.1233, 0.1385]	No	[-0.0792, 0.0928]	No	0.001
Day of Year	0.0183	0.0627	[-0.1029, 0.1455]	No	[-0.0629, 0.1017]	No	0.001
SHW							
MAX	-0.0244	0.0607	[-0.1095, 0.0990]	No	[-0.1099, 0.0549]	No	0.001
Year	0.0111	0.0664	[-0.1187, 0.1410]	No	[-0.0731, 0.0967]	No	0.001
Day of Year	0.0213	0.0635	[-0.0974, 0.1491]	No	[-0.0585, 0.1004]	No	0.001
Tx Area Alloted							
AHW							
MAX	0.0566	0.0861	[-0.1111, 0.2250]	No	[-0.0531, 0.1607]	No	0.004
Year	0.0809	0.0863	[-0.0784, 0.2434]	No	[-0.0243, 0.1868]	No	0.004
Day of Year	0.0059	0.0808	[-0.1787, 0.1966]	No	[-0.0877, 0.1408]	No	0.004
CIT							
MAX	0.0065	0.0722	[-0.1332, 0.1442]	No	[-0.0858, 0.1002]	No	0.003
Year	0.1102	0.0738	[-0.0404, 0.3472]	No	[0.0197, 0.2003]	Yes	0.003
Day of Year	0.0059	0.0688	[-0.1225, 0.1410]	No	[-0.0811, 0.0954]	No	0.003
MAP							
MAX	-0.0167	0.0759	[-0.1738, 0.1364]	No	[-0.1153, 0.0813]	No	0.003
Year	0.1041	0.0749	[-0.0401, 0.2577]	No	[0.0091, 0.1989]	Yes	0.003
Day of Year	0.0152	0.0732	[-0.1063, 0.1550]	No	[-0.0804, 0.1045]	No	0.003
MAX							
MAX	-0.0396	0.0703	[-0.1719, 0.1026]	No	[-0.1246, 0.0551]	No	0.004
Year	0.1327	0.0742	[-0.0253, 0.3440]	No	[0.0219, 0.2179]	Yes	0.004
Day of Year	0.0102	0.0694	[-0.1198, 0.1512]	No	[-0.0779, 0.1007]	No	0.004
Rh							
Rh	-0.0250	0.0663	[-0.1559, 0.1002]	No	[-0.1191, 0.0591]	No	0.004
Year	0.1148	0.0736	[-0.0244, 0.3246]	No	[0.0242, 0.2046]	Yes	0.004
Day of Year	0.0066	0.0681	[-0.1214, 0.1446]	No	[-0.0791, 0.0963]	No	0.004
Specialized Frequency							
AHW							
MAX	-0.0754	0.1307	[-0.1882, 0.3261]	No	[-0.0977, 0.2410]	No	0.027
Year	0.0171	0.1234	[-0.2056, 0.2795]	No	[-0.1477, 0.1798]	No	0.027
Day of Year	0.0842	0.1209	[-0.1707, 0.3071]	No	[-0.0826, 0.2249]	No	0.027
CIT							
MAX	-0.0034	0.1141	[-0.2242, 0.2190]	No	[-0.1485, 0.1411]	No	0.027
Year	0.0563	0.1116	[-0.1762, 0.2808]	No	[-0.0876, 0.2024]	No	0.027
Day of Year	0.0353	0.1065	[-0.1728, 0.2449]	No	[-0.1005, 0.1907]	No	0.027
MAX							
MAX	0.1099	0.1071	[-0.0950, 0.3344]	No	[-0.0209, 0.2488]	No	0.028
Year	0.0258	0.1076	[-0.1901, 0.2210]	No	[-0.1119, 0.1616]	No	0.028
Day of Year	0.0192	0.1055	[-0.1937, 0.2255]	No	[-0.1139, 0.1548]	No	0.028
MAP							
MXP	-0.1087	0.1089	[-0.3324, 0.1071]	No	[-0.2464, 0.0286]	No	0.028
Year	0.0562	0.1118	[-0.1984, 0.2987]	No	[-0.1096, 0.2087]	No	0.028
Day of Year	0.0540	0.1079	[-0.1614, 0.2811]	No	[-0.0843, 0.1960]	No	0.028
SHW							
MAX	-0.0253	0.1068	[-0.2389, 0.1909]	No	[-0.1437, 0.1072]	No	0.028
Year	0.0583	0.1097	[-0.1604, 0.2762]	No	[-0.0814, 0.1950]	No	0.028
Day of Year	0.0362	0.1049	[-0.1634, 0.2453]	No	[-0.0981, 0.1713]	No	0.028
Specialized Richness							
AHW							
MAX	-0.3206	0.1512	[-0.6782, 0.0158]	No	[0.6310, 0.0120]	Yes	0.228
Year	-0.1177	0.1546	[-0.4270, 0.1777]	No	[-0.3156, 0.0781]	No	0.228
Day of Year	0.1812	0.1446	[-0.1585, 0.4765]	No	[-0.0028, 0.3829]	No	0.228
CIT							
MAX	0.0154	0.1419	[-0.2984, 0.3090]	No	[-0.1629, 0.1950]	No	0.231
Year	-0.0887	0.1474	[-0.3844, 0.2356]	No	[-0.1974, 0.1214]	No	0.231
Day of Year	0.0898	0.1363	[-0.1767, 0.3393]	No	[-0.0819, 0.2641]	No	0.231
MAX							
MAX	0.1583	0.1299	[-0.1529, 0.3632]	No	[-0.0631, 0.3304]	No	0.232
Year	-0.0312	0.1458	[-0.3253, 0.2512]	No	[-0.2739, 0.1535]	No	0.232
Day of Year	0.0778	0.1366	[-0.1929, 0.3513]	No	[-0.0840, 0.2507]	No	0.232
Rh							
Rh	-0.1774	0.1264	[-0.4407, 0.0754]	No	[-0.3381, -0.0060]	Yes	0.231
Year	0.0124	0.1368	[-0.2697, 0.2795]	No	[-0.1636, 0.1873]	No	0.231
Day of Year	0.0992	0.1335	[-0.1665, 0.3969]	No	[-0.0695, 0.2772]	No	0.231
SHW							
MAX	0.0234	0.1399	[-0.2486, 0.3940]	No	[-0.1528, 0.2017]	No	0.233
Year	-0.0117	0.1432	[-0.3046, 0.2580]	No	[-0.1986, 0.1755]	No	0.233
Day of Year	0.0801	0.1357	[-0.1711, 0.3967]	No	[-0.0874, 0.2627]	No	0.233
All Frequency							
AHW							
MAX	-0.0041	0.1432	[-0.3003, 0.2259]	No	[-0.2039, 0.1501]	No	0.002
Year	0.0163	0.1283	[-0.2147, 0.3370]	No	[-0.1899, 0.2401]	No	0.002
Day of Year	0.0038	0.1245	[-0.1833, 0.3101]	No	[-0.1874, 0.2217]	No	0.002
CIT							
MAX	-0.0471	0.1131	[-0.2809, 0.1779]	No	[-0.1924, 0.0981]	No	0.002
Year	0.0750	0.1171	[-0.1537, 0.3369]	No	[-0.0712, 0.2242]	No	0.002
Day of Year	0.0814	0.1055	[-0.1128, 0.2879]	No	[-0.0524, 0.2018]	No	0.002
MAX							
MAX	0.1749	0.1586	[-0.0903, 0.5393]	No	[-0.0318, 0.3577]	No	0.004
Year	0.0237	0.1136	[-0.2071, 0.2634]	No	[-0.1207, 0.1672]	No	0.004
Day of Year	0.0846	0.1076	[-0.1463, 0.3156]	No	[-0.0713, 0.2814]	No	0.004
MAP							
MXP	-0.0654	0.1134	[-0.2909, 0.1596]	No	[-0.2046, 0.0763]	No	0.002
Year	0.0435	0.1119	[-0.1756, 0.2616]	No	[-0.1065, 0.1935]	No	0.002
Day of Year	0.0960	0.1059	[-0.1241, 0.3069]	No	[-0.0501, 0.2286]	No	0.002
SHW							
MAX	-0.0403	0.1081	[-0.2531, 0.1979]	No	[-0.1795, 0.0958]	No	0.002
Year	0.0619	0.1099	[-0.1589, 0.2774]	No	[-0.0791, 0.2011]	No	0.002
Day of Year	0.0813	0.1051	[-0.1205, 0.2906]	No	[-0.0541, 0.2147]	No	0.002
All Richness							
AHW							
MAX	0.1571	0.1495	[-0.1304, 0.4647]	No	[-0.0326, 0.3439]	No	0.284
Year	0.0584	0.1563	[-0.3663, 0.2712]	No	[-0.2241, 0.1693]	No	0.284
Day of Year	0.2758	0.1492	[-0.0763, 0.5122]	No	[0.0274, 0.4956]	Yes	0.284
CIT							
MAX	-0.0073	0.1861	[-0.3776, 0.3577]	No	[-0.1796, 0.1679]	No	0.286
Year	0.0502	0.1417	[-0.2243, 0.3303]	No	[-0.1306, 0.2311]	No	0.286
Day of Year	0.1557	0.1338	[-0.0986, 0.4459]	No	[-0.0124, 0.3382]	No	0.286
MAX							
MAX	0.1969	0.1577	[-0.1013, 0.5633]	No	[-0.0355, 0.3651]	No	0.287
Year	0.0181	0.1437	[-0.2682, 0.2992]	No	[-0.1635, 0.1977]	No	0.287
Day of Year	0.1383	0.1351	[-0.1228, 0.4862]	No	[0.0353, 0.3505]	Yes	0.287
Rh							
Rh	-0.1864	0.1237	[-0.3907, 0.0993]	No	[-0.2927, 0.0183]	No	0.285
Year	0.0634	0.1392	[-0.1969, 0.3287]	No	[-0.1092, 0.2329]	No	0.285
Day of Year	0.1618	0.1360	[-0.0871, 0.4076]	No	[-0.0499, 0.3551]	No	0.285
Tp							
TD	0.3486	0.1181	[0.0163, 0.6808]	Yes	[0.0979, 0.5962]	Yes	0.351
Year	0.0190	0.1262	[-0.2348, 0.3430]	No	[-0.1426, 0.1796]	No	0.351
Day of Year	0.1598	0.1262	[-0.0865, 0.4164]	No	[0.0007, 0.3214]	Yes	0.351

Figure S1:

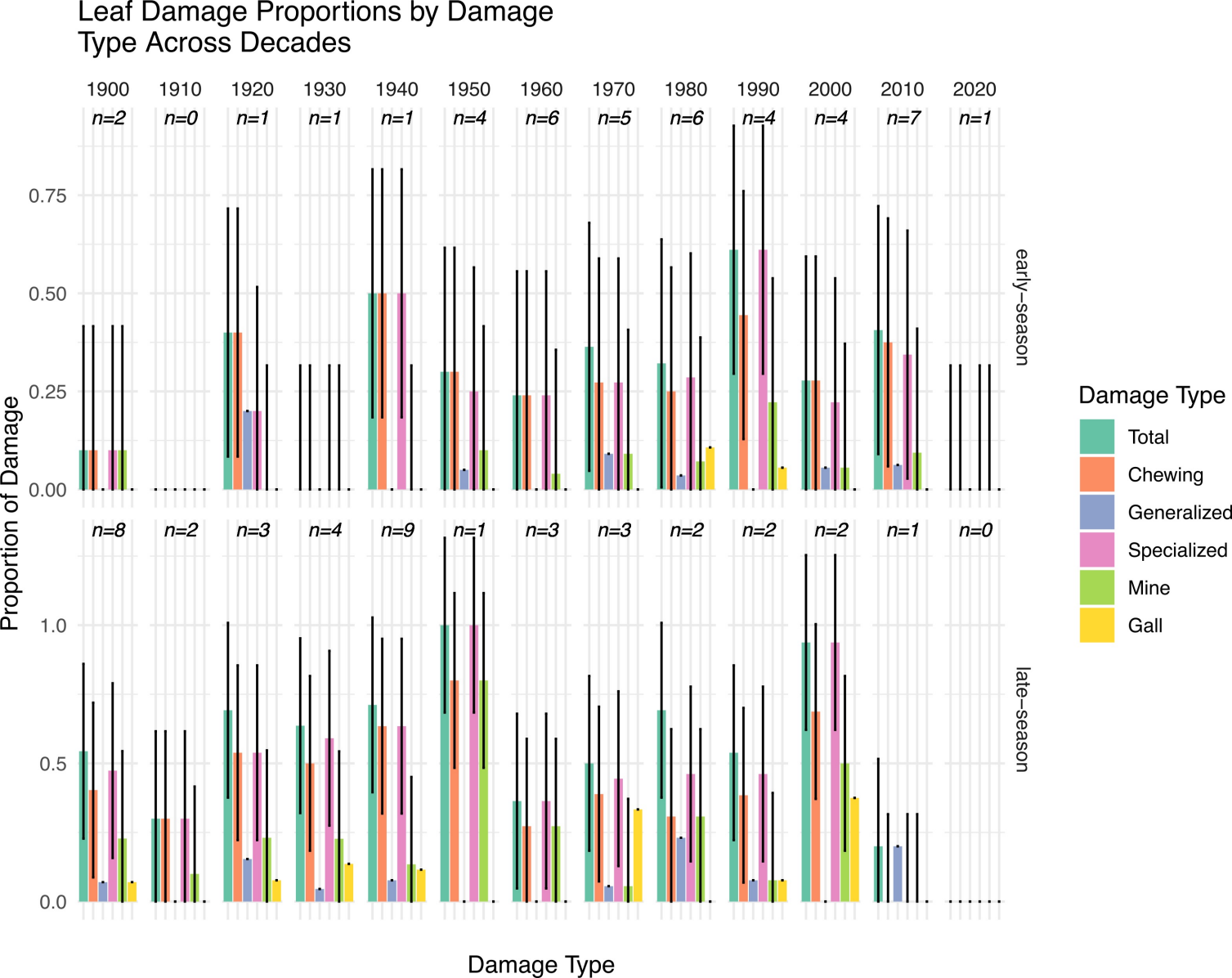
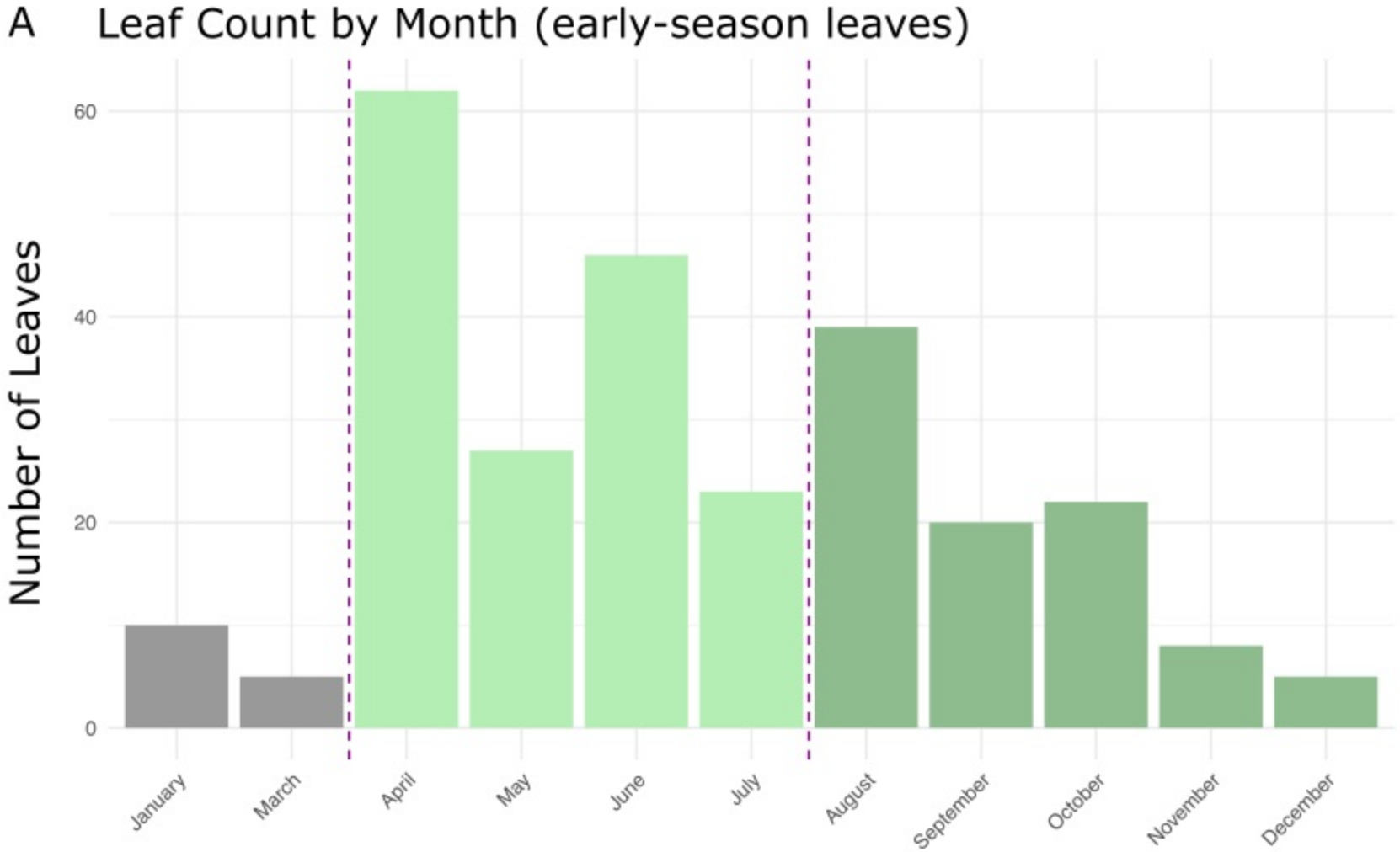

Leaf Damage Proportions by Damage Type Across Decades
1900
1910
1920
1930
1940
1950
1960
1970
1980
1990
2000
2010
2020
n=2
n=0
n=1
n=1
n=1
n=4
n=6
n=5
n=6
n=4
n=4
n=7
n=1
n=8
n=2
n=3
n=4
n=9
n=1
n=3
n=3
n=2
n=2
n=2
n=1
n=0
0.75
0.50
0.25
0.00
1.0
0.5
0.0
Proportion of Damage
early−season
late−season
Damage Type
Total
Chewing
Generalized
Specialized
Mine
Gall
Damage Type

Figure S2:


A
Leaf Count by Month (early-season leaves)
Number of Leaves
60
40
20
0
January
March
April
May
June
July
August
September
October
November
December

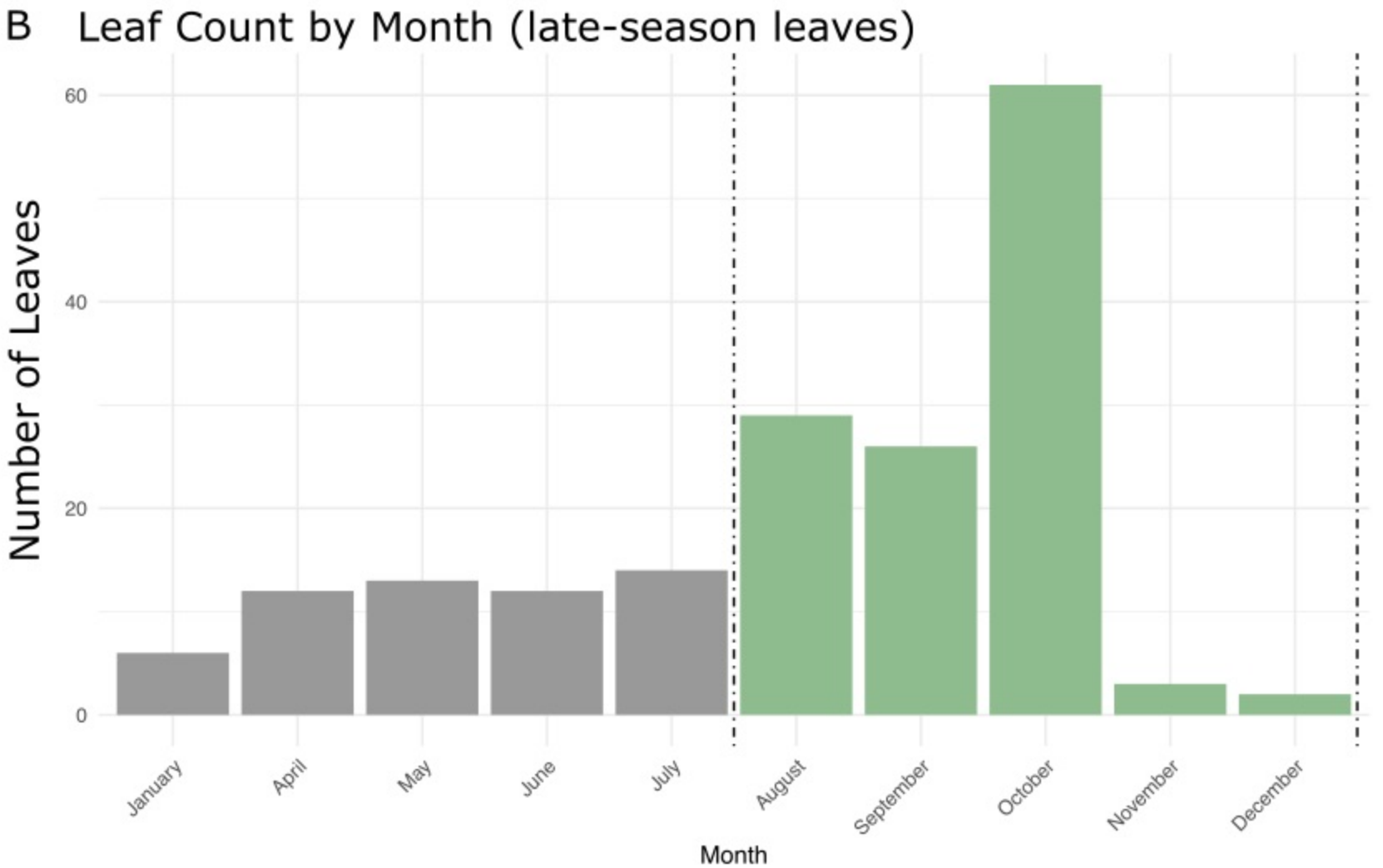
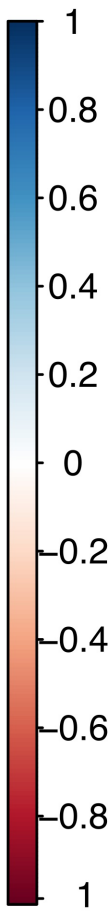
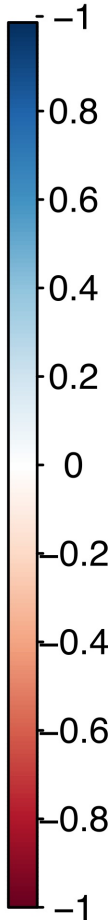

B
Leaf Count by Month (late-season leaves)
Number of Leaves
60
40
20
0
January
April
May
June
July
August
September
October
November
December
Month

Figure S3:

A

	Year	Month	Day	DOY	Latitude	Longitude
Year	1.00	−0.11	[illegible]	−0.11	0.17	−0.12
Month	−0.11	1.00	−0.02	0.97	[illegible]	−0.08
Day	[illegible]	−0.02	1.00	0.21	−0.13	0.16
DOY	−0.11	0.97	0.21	1.00	−0.03	−0.05
Latitude	0.17	[illegible]	−0.13	−0.03	1.00	−0.83
Longitude	−0.12	−0.08	0.16	−0.05	−0.83	1.00

B

	Year	Month	Day	DOY	Latitude	Longitude
Year	1.00	0.16	0.13	0.17	−0.45	0.51
Month	0.16	1.00	−0.10	0.99	−0.15	0.13
Day	0.13	−0.10	1.00	0.02	0.21	−0.12
DOY	0.17	0.99	0.02	1.00	−0.13	0.12
Latitude	−0.45	−0.15	0.21	−0.13	1.00	−0.83
Longitude	0.51	0.13	−0.12	0.12	−0.83	1.00